



Investment Society
Imperial College London



Lecture 5

Derivatives II

Securities and Education Certificate

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Slide deck inspired by Dr. Euan Doidge and Prof. Anthony Barrett (Department of Chemistry, Imperial College London)

Slide designs by Ahmad Danesh (Head of Editorial, Imperial College Investment Society 2021/22)

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Background

Some context to introduce you to a concept

Fact

You will need to recall this information in the exam

Model Example

A fully detailed breakdown of an example (question)

Try it yourself

It's your turn to give this problem a go and test your knowledge!

Equation sheet

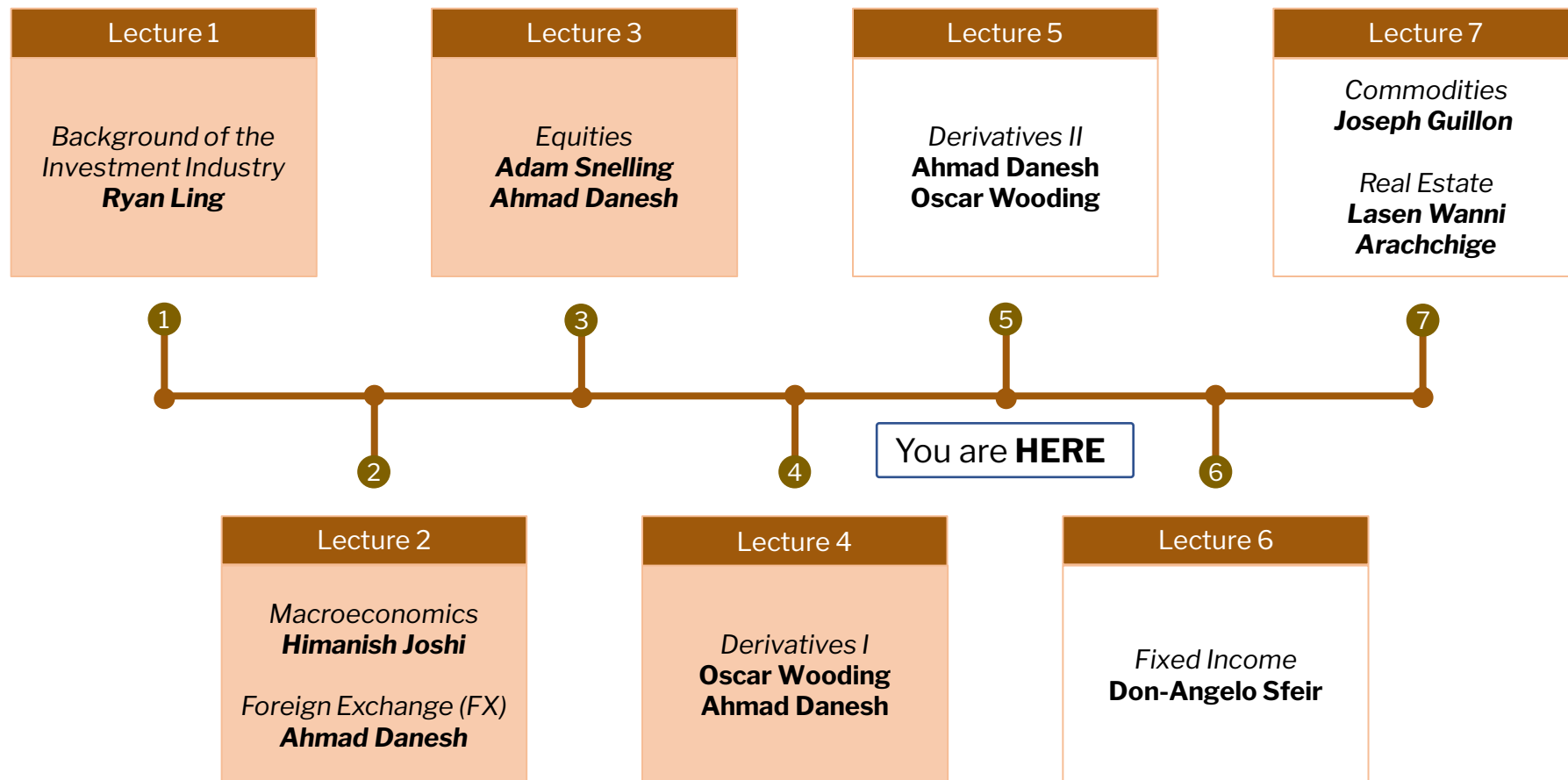
Any equations that you may need for the exam are provided

Taking it further

(non-examinable) stuff to show you where this concept can be applied

The slides are colour-coded to aid your understanding of the course and ensure that you are familiar with the demands of the content provided in each slide.

Lecture Timeline



Futures and Forwards (recap)

Let's start with the hidden time machine invented for grain farmers in 1848!

☞ Both futures and forwards contracts allow investors to do 3 things:

Buy OR sell a fixed amount of an underlying asset

At a fixed price

☞ Common underlying assets:
currencies, commodities and bonds.

☞ The quoted price → how much **must** be
exchanged at expiry.

At a fixed date in the future

☞ This is the **expiry** or **settlement date**.

☞ The date → when the initially agreed price is
exchanged for the fixed amount of asset.

☞ **NOTE** these contracts create an **obligation** between the two parties to exchange assets.

Forwards - Understanding a Fair Price

What objectives does a fair price have to meet?

Banks

☞ Create a contract with no market risk.

☞ Use *borrowing* and *lending* to maximal benefit.

☞ Avoid use of the bank's capital.

Clients

☞ Spot and forwards should be equally attractive in terms of payoff over the contract period

☞ Longing and Shorting should be equally attractive.

Forwards are **ALWAYS** priced as the future value of the underlying asset.

Pricing FX Forwards

A **Foreign Exchange forward** is an agreement to **exchange a set amount of one currency with another at a predetermined exchange rate** on a **set time in the future**.

You can also get FX futures but these are priced through supply and demand.

Let's try an example to see how these contracts work from a bank's perspective.

We are a bank and our client wants to buy 1,000,000 GBP against USD. (N.B. 'against' = GBP/USD)

Spot GBP/USD is quoted at: £1 = \$1.18.

But our customer requires delivery 2 years after the spot date.

What forward FX rate do we quote?

Pricing FX Forwards

What steps would a bank take to make this contract risk free for them?

To be able to offer this contract to clients, a bank must be able to make it with no risk.

They don't want to use any of their own resources, only the available markets (lending, borrowing, ...).

👉 As soon as the contract is formed, this is what a bank may do...

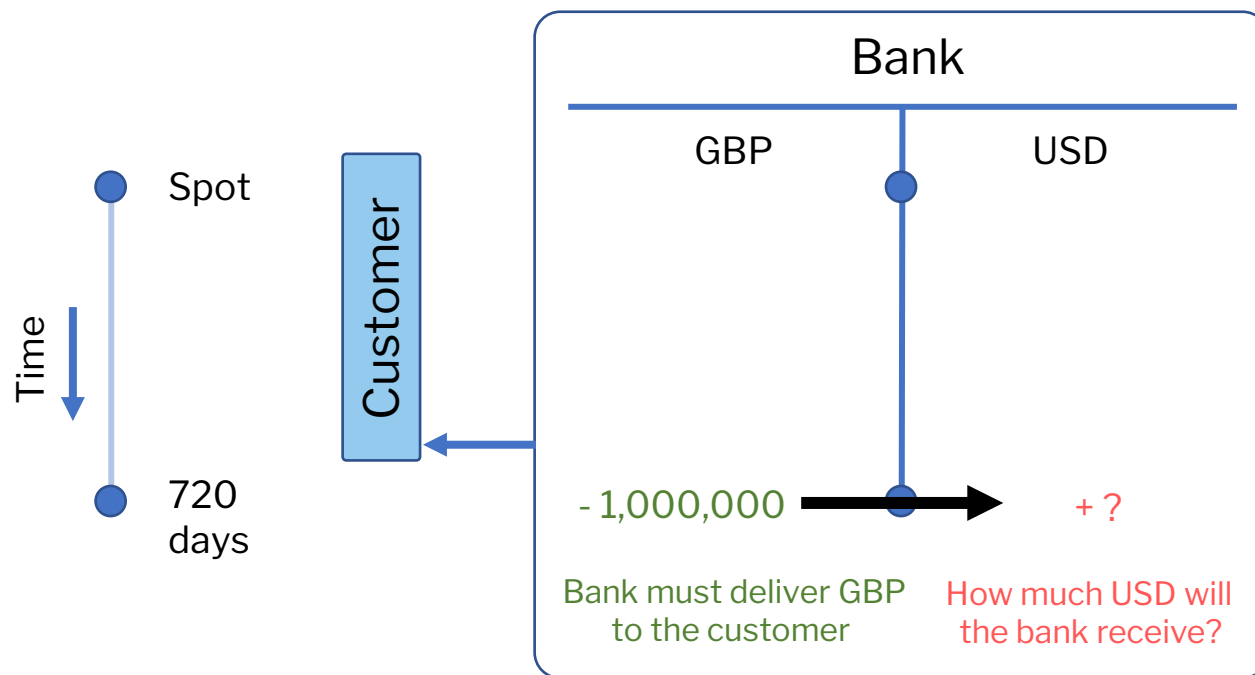
- 1. Borrow the currency to be received at expiry**, discounted by interest over the contract period.
- 2. Convert the currency at today's spot price.**
- 3. Lend the converted cash and gain interest** until expiry where it is given to the customer.

Pricing FX Forwards

Mapping the cashflows can be a useful technique in pricing

Spot GBP/USD is quoted at: 1.18 (£1 = \$1.18).

Customer wants 1,000,000 GBP against USD.



☞ Time is plotted going down the page.

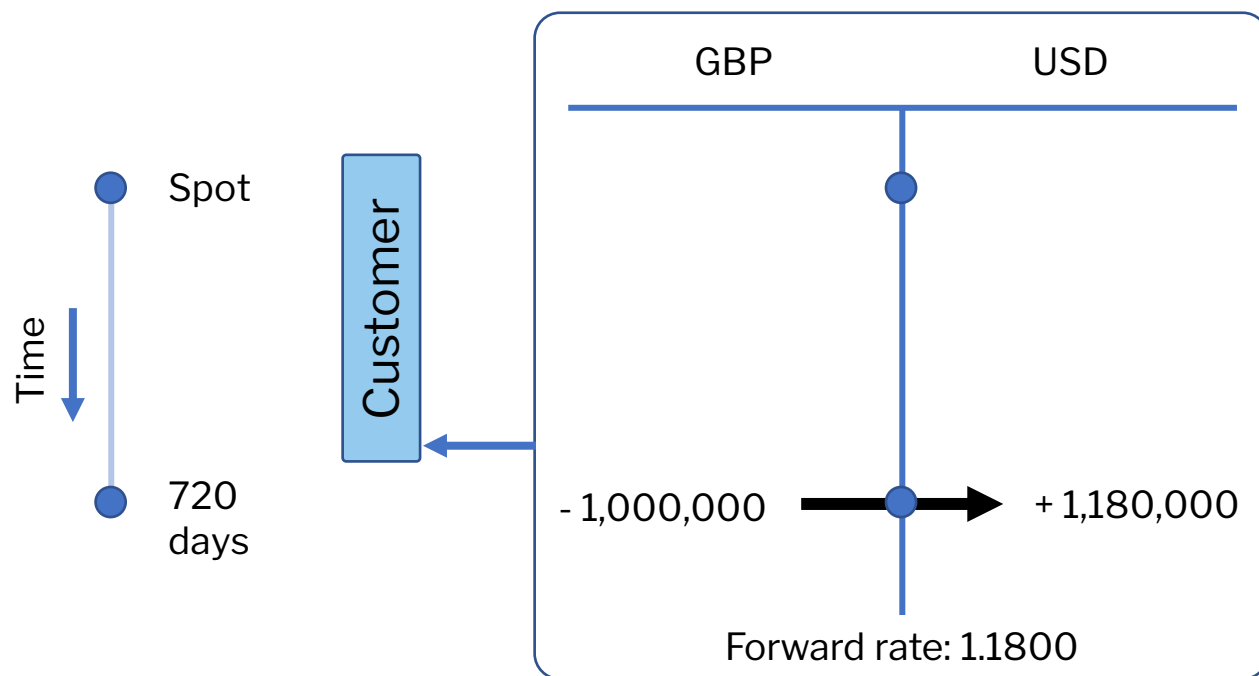
Pricing FX Forwards

Scenario 1: Forward price is equal to the spot price

Spot GBP/EUR is quoted at: 1.18 (£1 = \$1.18).

360 day GBP interest rate: 1.75% (Aug 2022)

360 day USD interest rate: 2.50% (Aug 2022)

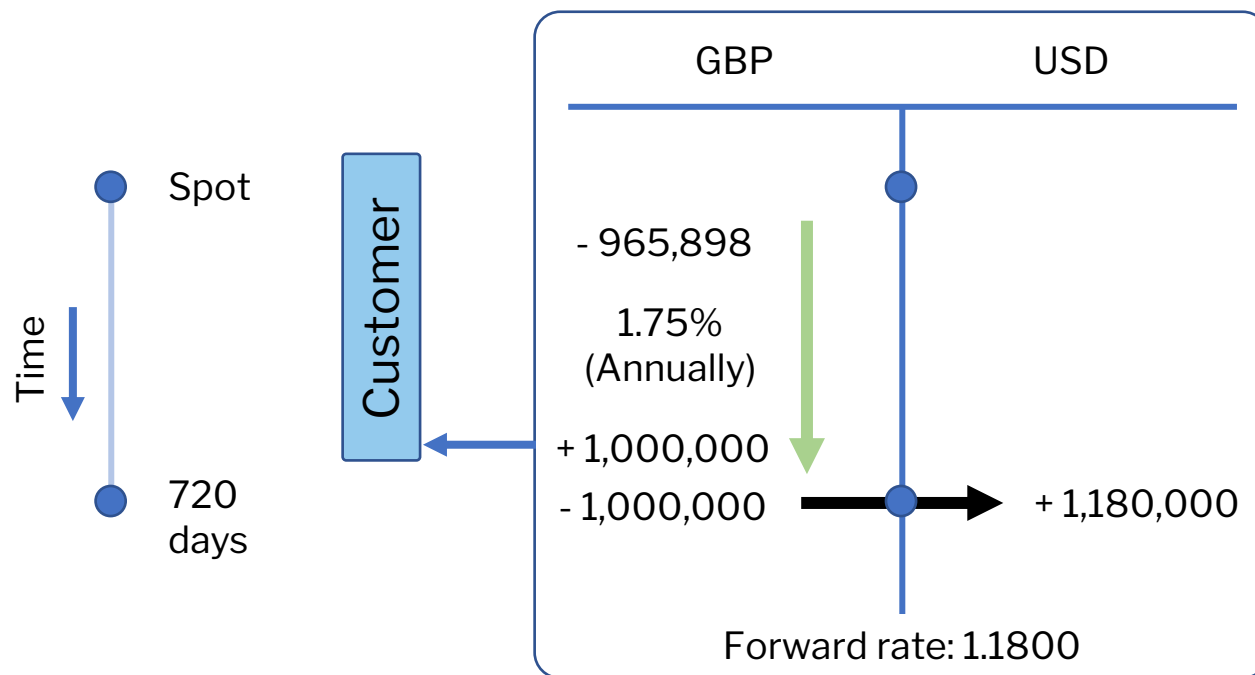


Pricing FX Forwards

Scenario 1: Forward price is equal to the spot price

What can the bank do immediately after entering the contract to hedge all risk?

1. Lend the currency sold forward.

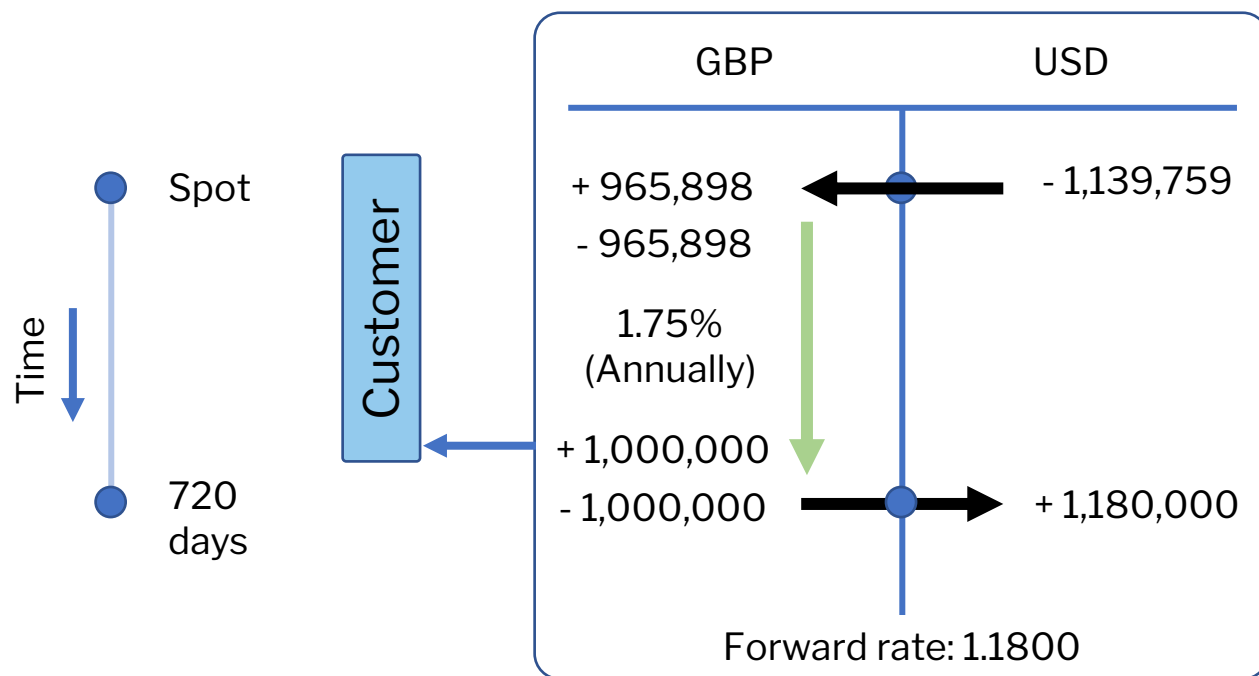


Pricing FX Forwards

Scenario 1: Forward price is equal to the spot price

2. Execute a spot deal.

Now we know how many dollars we need today to guarantee the client is paid enough in 2 years

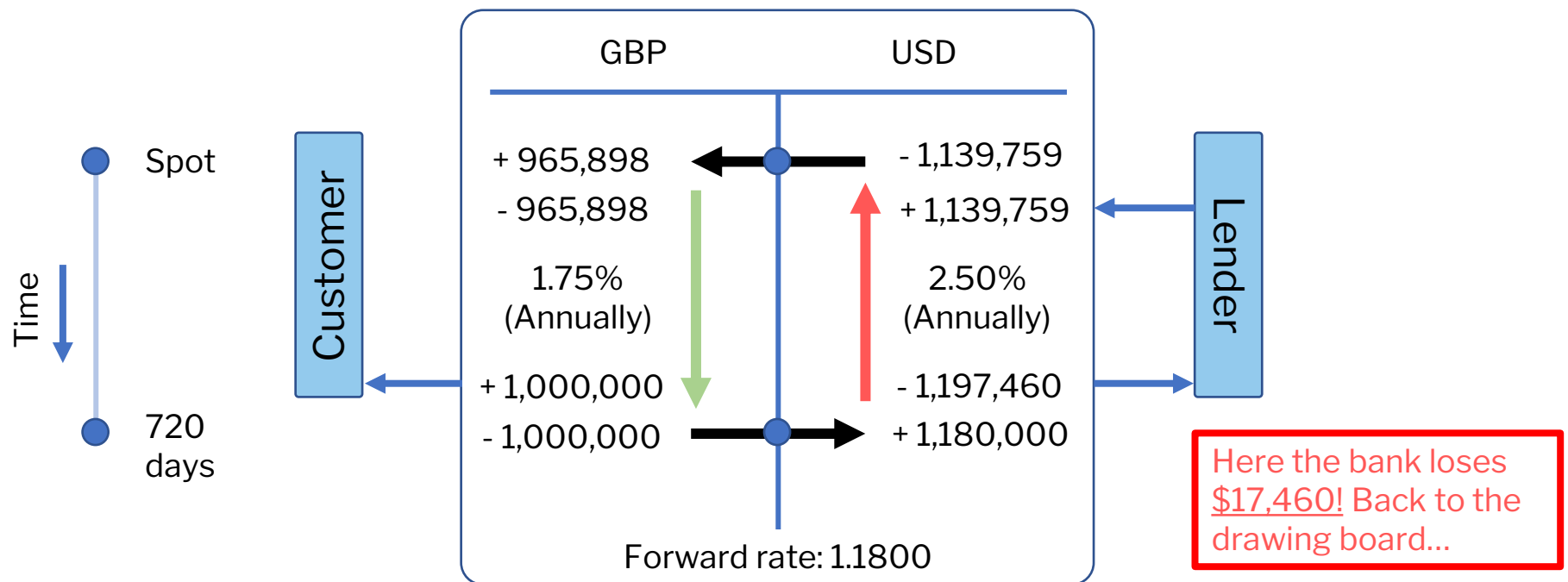


Pricing FX Forwards

Scenario 1: Forward rate is equal to the spot rate

3. Borrow the currency bought forward

👉 To avoid using firm capital, the bank borrows this amount of dollars until the expiry date



Pricing FX Forwards

So... what went wrong with this deal?

The current spot rate could be the forward rate if interest rates are the **SAME** in each currency.

In most cases, the lending and borrowing rates won't be equal and so cash flows don't match.

To avoid losing money, this forward rate loop can be solved by considering the following equations.

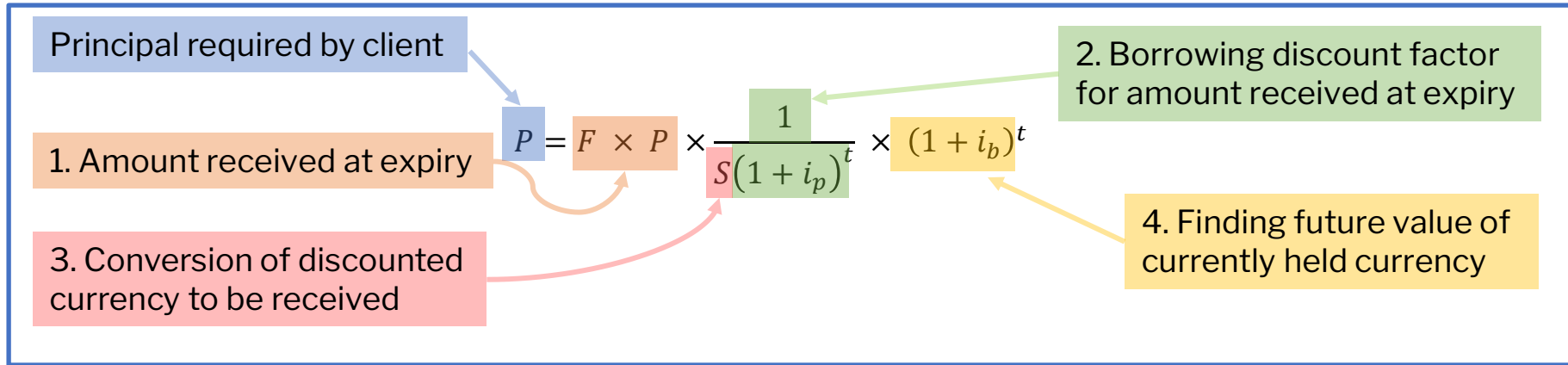
$$P = F \times P \times \frac{1}{S(1 + i_p)^t} \times (1 + i_b)^t$$

$$F = \frac{(1 + i_p)^t}{(1 + i_b)^t} \times S$$

P = client principal, F = forward rate, S = spot rate, i_b = interest on base, i_p = interest on pricing, t = time

Pricing FX Forwards

This equation can be broken down into each stage of the trade.



$$F = \frac{(1+i_p)^t}{(1+i_b)^t} \times S \quad (1)$$

$$F = \frac{(1+i_p \times \frac{t}{B})}{(1+i_b \times \frac{t}{B})} \times S \quad (2)$$

(1) – For expiries over a year away (t is in years)

(2) – expiry under a year away (t is in days, B = day count basis)

P = client principal (In the currency that client is receiving)

F = forward rate (With the currency received by the client as base)

S = spot rate (With the currency received by the client as base)

i_p = interest on pricing currency (2nd one in pair)

i_b = interest on base currency (1st one in pair)

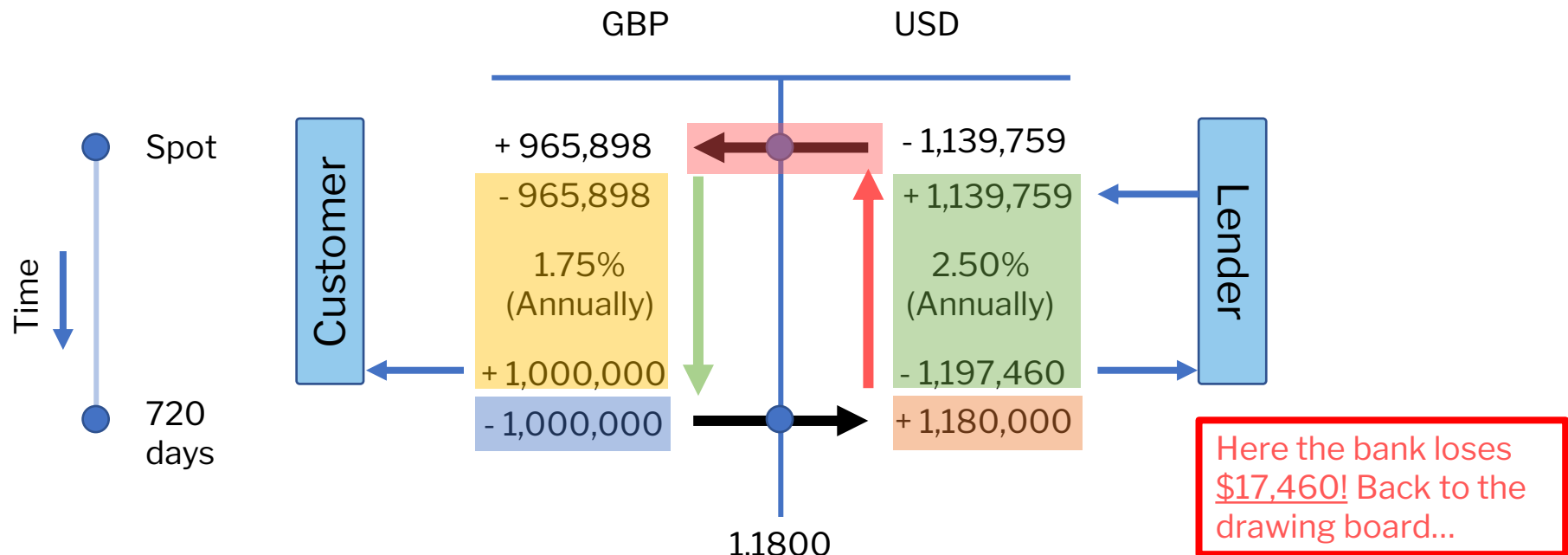
t = time (time until contract expiry, eq. (1)-years, eq. (2)-days)

Pricing FX Forwards

Scenario 1: Forward price is equal to the spot price.

3. Borrow the currency bought forward

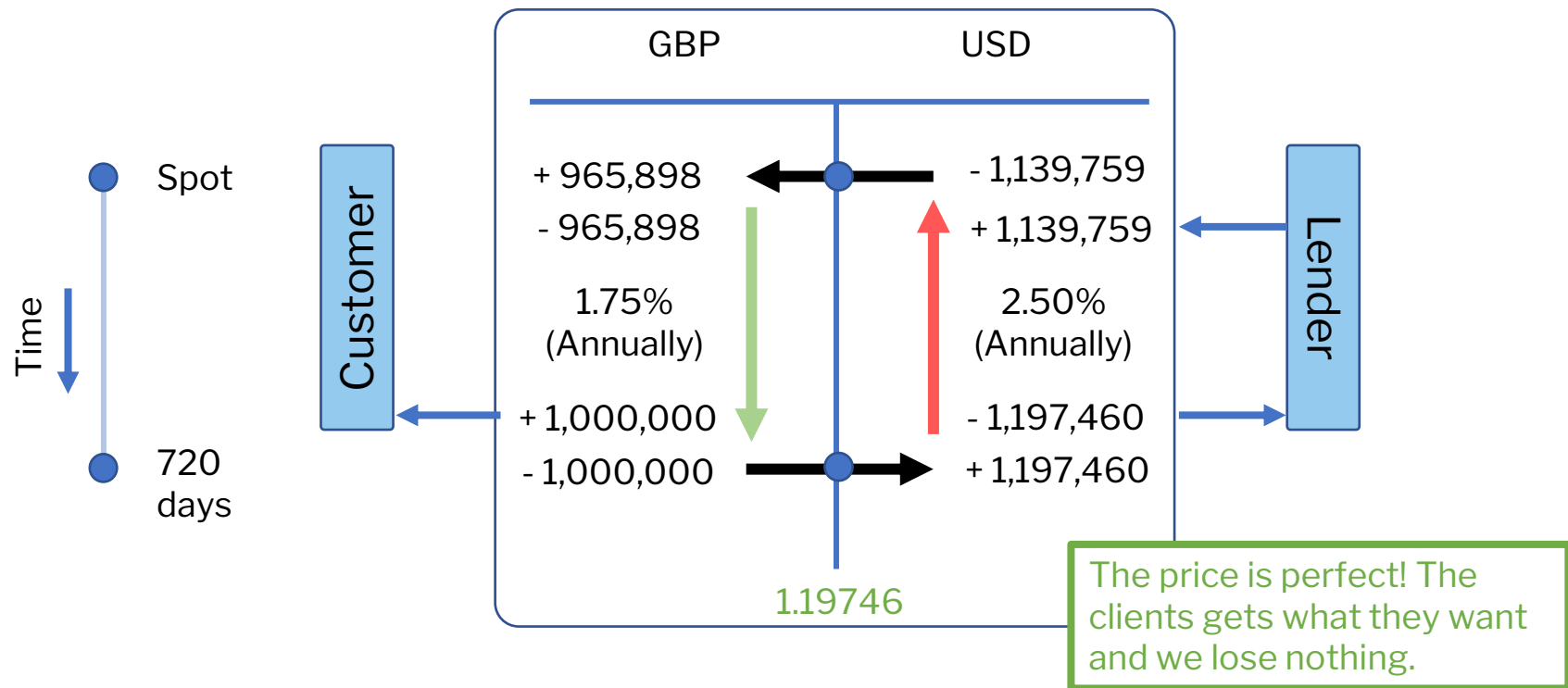
👉 To avoid using firm capital, the bank borrows this amount of dollars until the expiry date



Pricing FX Forwards

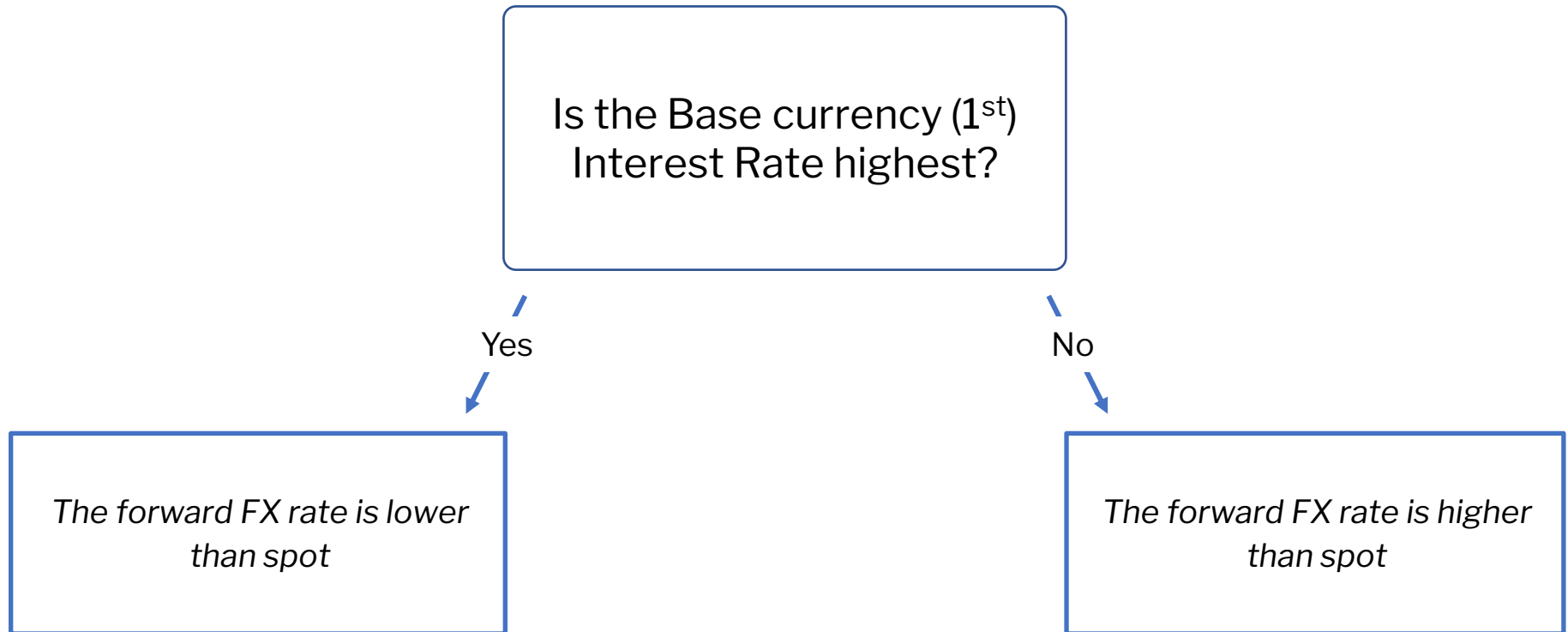
Scenario 2: Forward price is adjusted for uneven interest rates.

$$F = \frac{(1 + 2.5\%)^2}{(1 + 1.75\%)^2} \times 1.18 = 1.19746$$



Prices vs Interest Rates

This flowchart demonstrates a quick rule to help check if your price is right.



The larger the difference in the two rates, the further away from the spot the forward price is.

Prices over Time

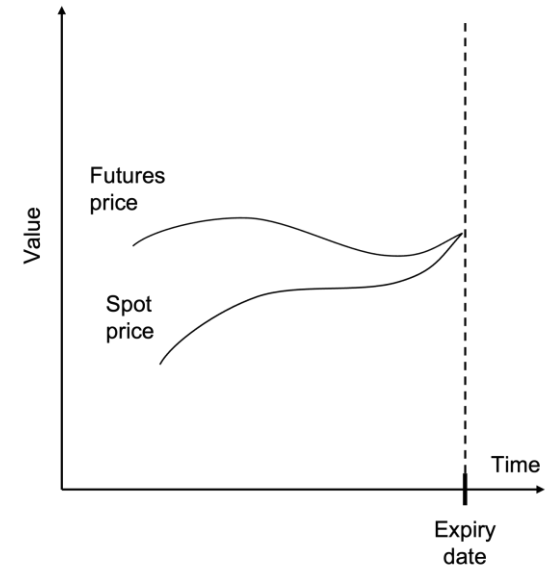
What happens to the price of these contracts as the expiry date nears?

Let's look at our FX equation again ...

$$f(t) = \frac{(1 + i_p)^t}{(1 + i_b)^t} \times S$$

$$f(t) = \frac{(1 + i_p \times \frac{t}{B})}{(1 + i_b \times \frac{t}{B})} \times S$$

$$\lim_{t \rightarrow 0} f(t) = S$$



👉 As the forward or futures contract approaches expiry, the price tends towards the underlying spot price.

Markets vs Fair Price

But Futures are exchange traded, so why would traders care about a 'fair' value?

👉 Despite futures market prices being driven by traders, the fair price is typically close to the quote price.

👉 This occurs due to **Efficient Market hypothesis** i.e. there can be no arbitrage opportunities

Efficient market hypothesis states that all information is reflected in markets, so **no arbitrage exists**

Arbitrage implies that there is riskless profit available – in reality, arbitrage is very difficult to come by!

E.g. buying PS5's for £499 on Game and selling them for £800 on eBay - riskless profit

What if the market price was higher than fair value? Let's look at an example with AMZN stock

Markets vs Fair Price

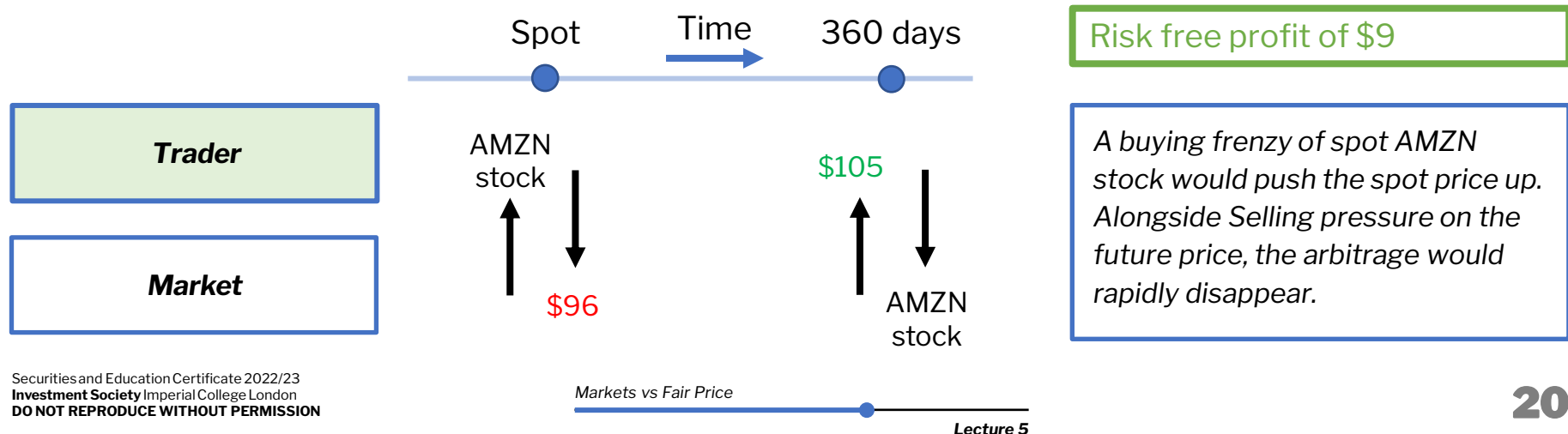
What arbitrage exists from an inflated market price, and how does the gap close?

Amazons shares currently trade at a spot price of \$96 with a dividend of 0% → no convenience yield.

Using our earlier formula, and a risk free rate of 5%, the 1 yr future value should be $\$96 \times 1.05 = \100.8

Scenario 1 – The 1yr future spot price is higher that fair value (\$105)

An arbitrage exists by **buying the asset now** and **selling the future**.



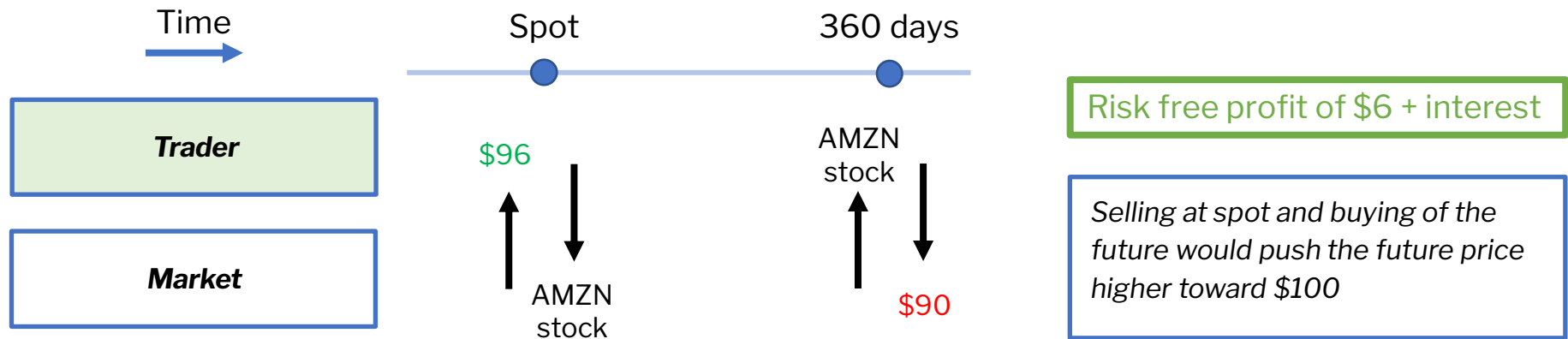
Markets vs Fair Price

Scenario 1 – The 1yr future spot price is higher than fair value (\$105)

The spot price would increase, and the future price would decrease until an equilibrium is formed.

Scenario 2 – The 1yr future spot price is higher than fair value (\$90)

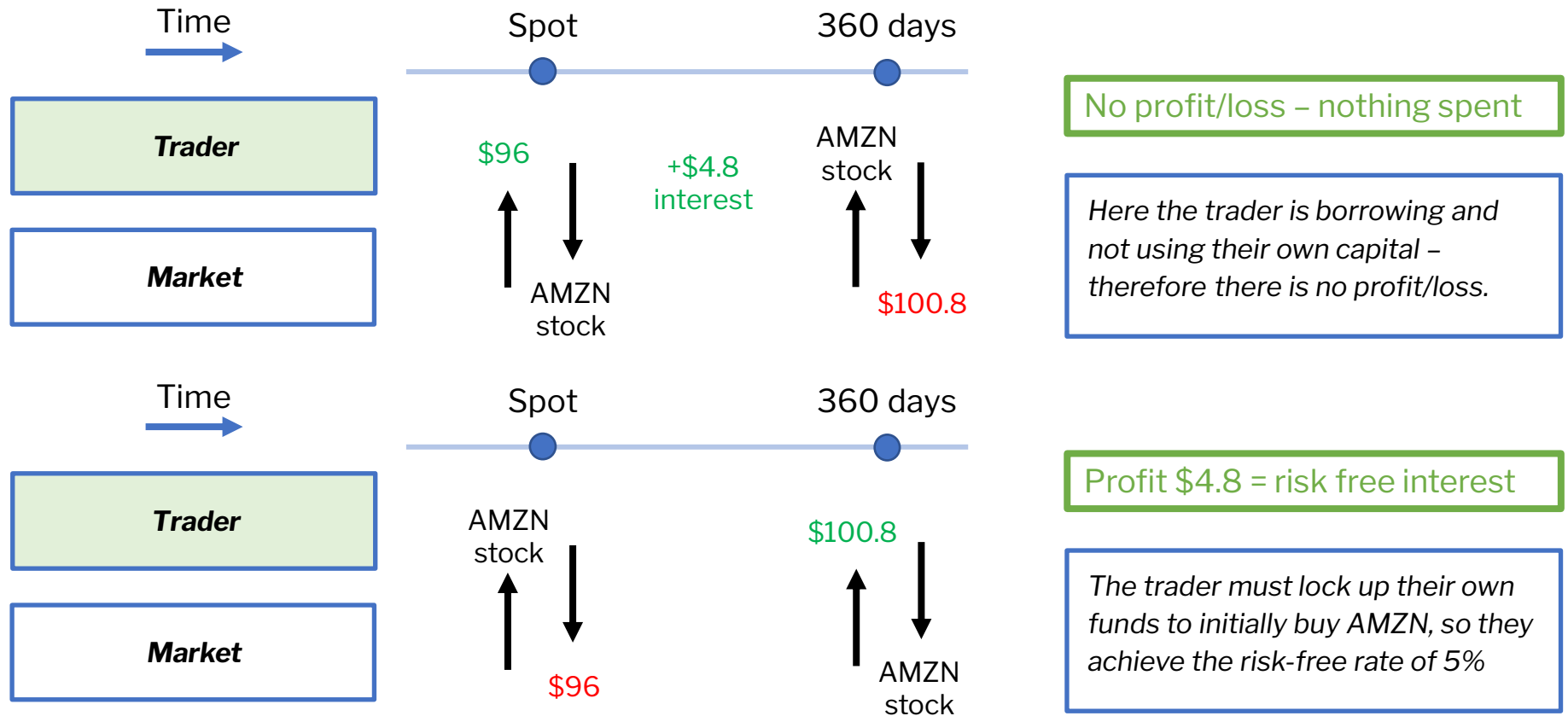
An arbitrage exists by **selling the asset now** and **buying the future**.



Markets vs Fair Price

Scenario 3 – The 1yr future spot price is at fair value

Fair value makes buying and selling equally attractive and removes any arbitrage opportunities.

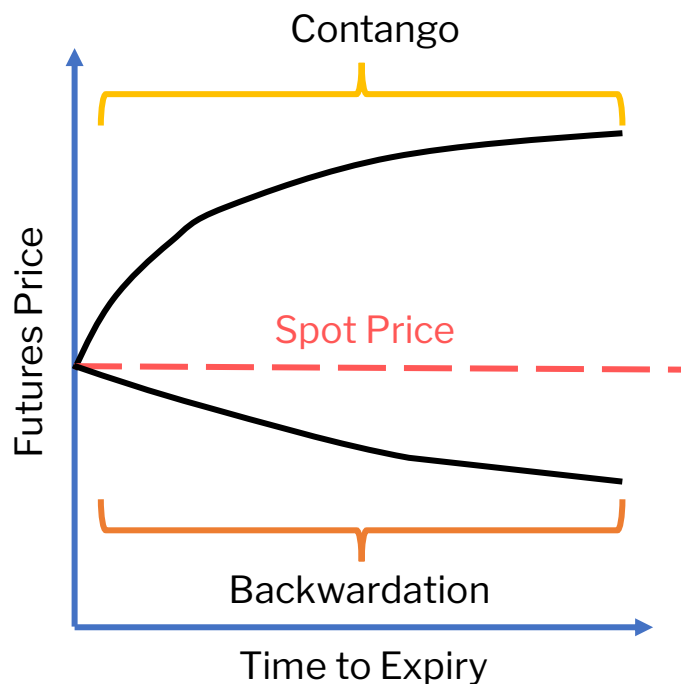


Forward/Futures Quotes Over Time

We can see this concept through forward curves

Forward curves show the price of different futures contracts with different expiry dates.

It is typical for forward curves to slope upward as the future value of an asset increases further away.



Reasons for **Contango**

Storage

Financing

Insurance

👉 The holder demands more to offset costs before expiry.

Backwardation - futures prices are decreasing with increasing time to expiry.
This could occur with very high demand for the spot asset now after e.g. a disaster.

Let's resume in 5 minutes...



Oscar Wooding

Head of Portfolio Management

Lead Analyst (2021 – 22)

Summer Analyst, Brevan Howard (2022)

Founder, QT Capital (2021 –)

Let's resume...



Ahmad Danesh
Lead Lecturer
Founder, QT Capital

Lecture 5

Derivatives II

Options

They're different to futures and forwards. How?

☞ Futures and forwards were *obligations* to trade an UNDERLYING asset (UL).

☞ The selling party must sell the UL at an agreed point on the future.

What if the derivative holder wants the *option* to trade a UL?

☞ **Option** contracts offers the right *but not the obligation* to buy or sell a UL at expiration.

☞ Options are derivatives, but are much more complex than futures and forwards.

Your notes

Types of option contracts

Why are option contracts useful?

☞ You can *choose* to exercise a contract → great for hedging risk!

☞ Two common types of option contracts are available:

Call options

Put options

☞ You **MUST** pay a cost to buy an options contract i.e. a '**premium**'

☞ Options are quoted per unit of UL, but are *standardised* to a specific quantity of UL.

Your notes

Call Options

Let's discuss the first type of options... Call options!

☞ Call options offer the buyer the right (but **NOT** the obligation) to **BUY** the UL:

Pre-determined strike price, K

Pre-determined (future) expiry date

How does a call option contract buyer benefit from a call option?

☞ Call options are beneficial when the UL market price S is greater than the strike price (i.e. $S > K$)

☞ If $S < K$, you buy the UL at the market price S and you *choose* **NOT** to exercise the call option.

Your notes

Call Options

Let's tie everything into a common example... buying stocks!

Rising oil prices mean that oil producers are likely to make record profits.

This is **NOT**
investment advice!

I want to buy shares in Saudi Aramco, trading at **35 SAR**.



Due to oil market uncertainty, I want a call option on Aramco's shares expiring 1st December.

On expiry, I think the share price will be **45 SAR**. The cost of an options contract is **5.00 SAR**.

Saudi Aramco Petrochemicals, Jubail, Saudi Arabia.
Suresh Babunair - Wikimedia Commons



Call Options

How do option contracts work for buying shares?

☞ An options premium costs 5 SAR per share.

☞ Options require purchase of **100** shares!

☞ I must pay $5 \text{ SAR} \times 100 \text{ shares} = 500 \text{ SAR}$ to enter 1 call option!

☞ The strike price is set at today's price, **35 SAR** (for illustrative purposes)

☞ On 1st December, if the share price = 45 SAR, I choose to *exercise* this call option.

☞ I pay $35 \text{ SAR} \times 100 = 3500 \text{ SAR}$ for 100 shares of Aramco.

Total Cost
 $= 3500 + 500 = \mathbf{4000 \text{ SAR.}}$

☞ I sell these 100 shares at the market price of 40.00 SAR, earning 4500 SAR (= 500 SAR profit).

If the share price went down, I'd make a loss. So, I do not exercise the options contract, **losing just 500 SAR.**

Put Options

If you understand Calls, you definitely understand Puts (they're the opposite!)

☞ Put options offer the buyer the right (but **NOT** the obligation) to **SELL** the UL:

Pre-determined strike price, K

Pre-determined (future) expiry date

How does a put option contract buyer benefit from a call option?

☞ Put options are beneficial when the UL market price S is lower than the strike price (i.e. $S < K$)

☞ If $S > K$, you sell the UL at the market price S and you *choose* **NOT** to exercise the put option.

Your notes

Put Options

Ideally, you'd be looking at a soon-to-be-failing business, so let's try football!

Juventus Football Club are having a terrible season and it looks to continue.

This is **NOT**
investment advice!

It currently trades at **0.34** EUR.

JUVENTUS

Due to uncertainty, I want a put option for my Juventus shares expiring 1st July.

On expiry, I think the share price will be **0.17** EUR. The cost of an options contract is **0.02** EUR.



Juventus vs Genoa (4-0), 23 April 2017
Save The Dream – Wikimedia Commons

Put Options

We can do the same analysis as we did for Juventus, but in the opposite way.

☞ I must pay $0.02 \text{ EUR} \times 100 \text{ shares} = \text{€}20$ to enter 1 put option.

☞ The strike price is set at today's price, **0.34 EUR**.

☞ On 1st July, if the share price is 0.17 EUR, I choose to *exercise* this call option.

☞ I sell my 100 shares of Juventus for $0.34 \text{ EUR} \times 100 = \text{€}34$

Total Profit
 $= 34 - 20 = \text{€}14$

☞ Without a put option, I would have sold 100 shares at €17.

Your notes

Intrinsic Value

Where does the idea of intrinsic value come from in the context of options?

👉 Consider a call option **WITHOUT** a premium.

Sony trades at \$90 at the NYSE. I want a call option with a 3 month expiry and a strike price of \$95

3 months later, the stock price reaches \$100. I get to pay \$95 for a stock that's worth \$100.

👉 The total profit *per share* is \$5.

👉 Therefore, the value of this call option is *up to* \$5.

👉 I am prepared to pay a **premium** up to \$5 for this option contract.

Your notes

Intrinsic Value

How do **call** option contracts relate to the idea of intrinsic value?

☞ The difference between the underlying asset price (S) and the strike price (K) → **intrinsic value** (IV)

$$IV = S - K$$

☞ A trader uses intrinsic value to price an options contract.

☞ In the example, the trader will try to set the premium as close to \$5 as possible.

☞ If the intrinsic value is zero or less (i.e. $S < K$), the contract is worthless.

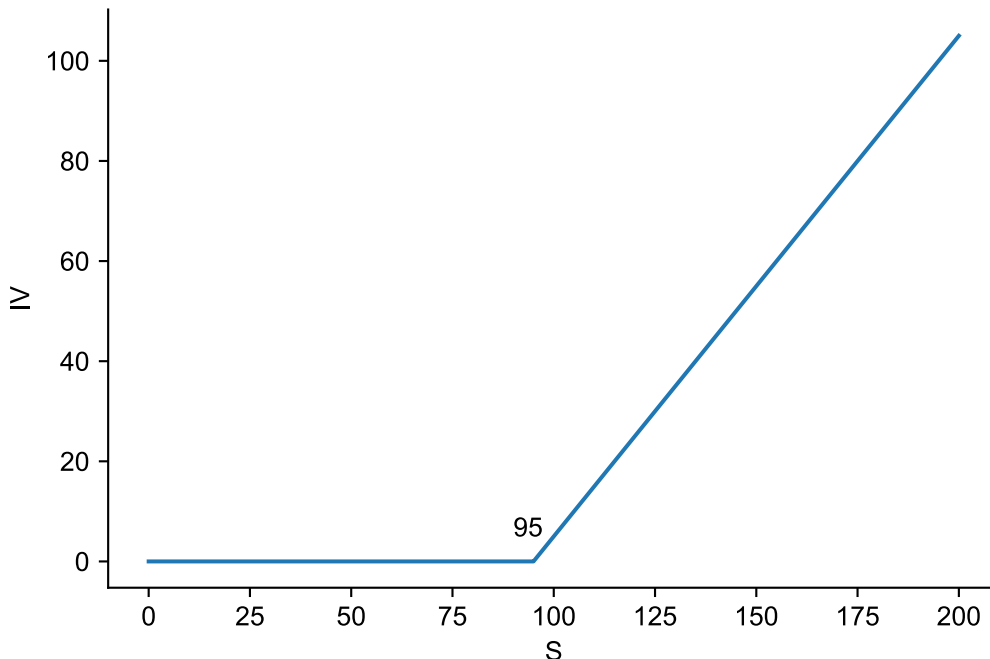
Your notes

Intrinsic Value

Piecewise functions are useful in modelling option contracts.

👉 We can use piecewise functions that give specific output values with conditions.

(Long) Call Option, Sony 95
Intrinsic Value



$$IV = \begin{cases} 0 & \text{if } S - K \leq 0 \\ S - K & \text{otherwise} \end{cases}$$

A linear graph when $S > 95$.
0 everywhere else.

👉 Alternative notations also exist for IV:

$$IV = \max(0, S - K)$$

$$IV = (S - K)^+$$

ITM, OTM & ATM

Some of you will be hearing the idea of *moneyiness*... What do we mean by that?

☞ If IV is positive for a given S, we say the option is **in-the-money (ITM)**.

$$IV = S - K$$

If a Sony 95 call option was exercised when the stock price was \$100, then the option is *in-the-money*.

☞ If IV is negative for a given S, we say the option is **out-the-money (OTM)** i.e. $S < K$.

☞ In some cases, IV can equal 0. Here, the option is **at-the-money (ATM)**.

☞ ITM does **NOT** mean profit!

☞ You need to account for the option premium!

Your notes

IV & Moneyness – Put Options

Everything you just read can be inverted for put options! An example will help.

Glencore trades on the LSE at 400 GBX (£4.00). A January 1 2023 400 Put option costs £0.90.

I think it'll go down to £3.00 on expiry because of difficulties in commodities. *Guess what, it does!*

👉 For a put option, intrinsic value is only achieved if $K > S$ so the equation is changed:

$$IV_{\text{put}} = K - S$$

Notice here that ($K, S = 4.00, 3.00$) so we want the stock price S to go down in order to get some value.

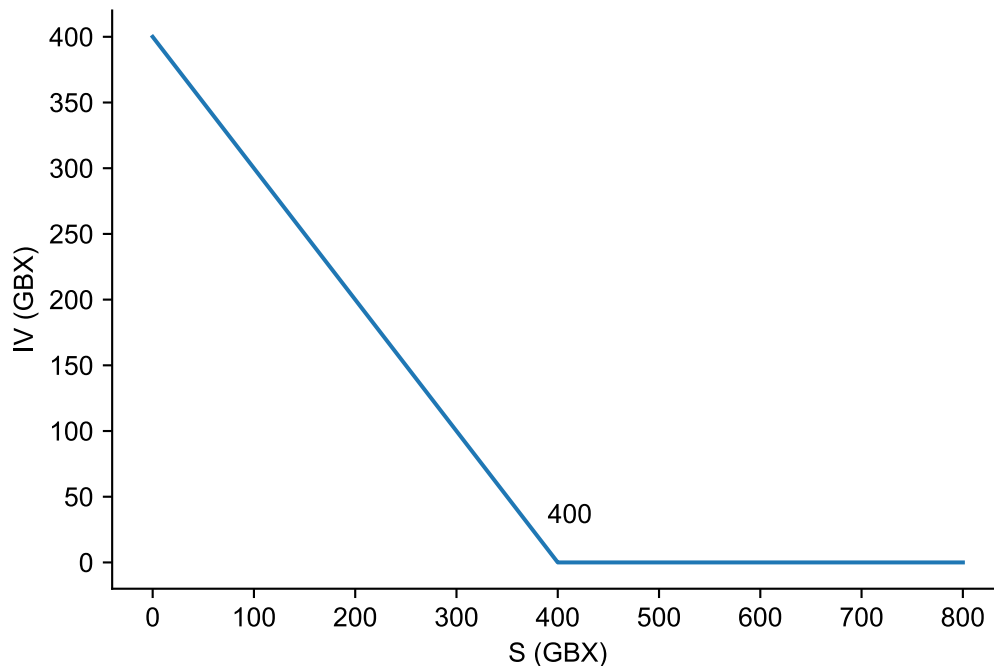
Your notes

Intrinsic Value

The graph for IV here is flipped horizontally – makes sense.

👉 Positive IV is achieved if the strike price K is lower the underlying share price S .

(Short) Put Option, Glencore 300 GBX
Intrinsic Value



$$IV = \begin{cases} K - S & \text{if } K - S \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

A linear graph when $S \leq 400$.
0 everywhere else.

👉 Alternative notations also exist for IV :

$$IV = \max(0, K - S)$$

$$IV = (K - S)^+$$

Payoff diagrams vs P&L graphs

I purposely hid the technical term for the IV graphs you just saw... Why?

👉 IV graphs are called **payoff diagrams**.
They show much money you get from an option at a given share price S .

👉 They are **NOT the same as profit and loss (P&L) graphs!**

For the Glencore 300 Call, if the share price reaches 200 GBX, the contract holder (me) is paid off £2.

Due to the option **premium**, I earn $£2 - £0.90 = £1.10$ profit.

👉 Payoff = £2.00, P&L = £1.10

Can you draw a graph that shows the P&L of entering the Glencore 300 call?

Your notes

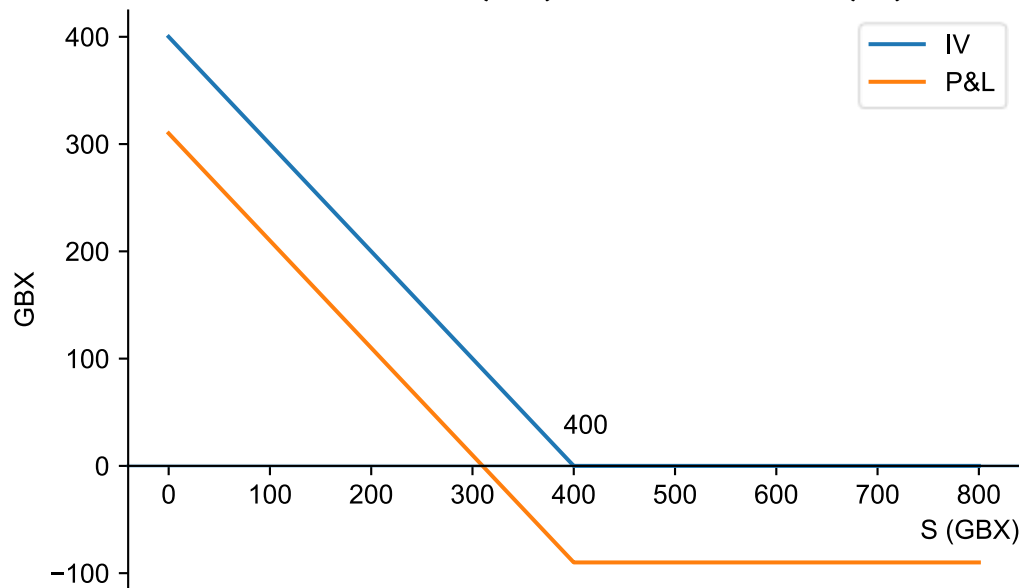
P&L graphs – Put Options

Some high school accounting might do the trick here...

👉 You know that profit = revenue – cost.

👉 Revenue = **payoff**, cost = **premium**.

(Short) Put Option, Glencore 300 GBX
Profit and Loss (P&L) vs. Intrinsic Value (IV)



$$\text{Profit} = \text{IV} - 0.90$$

$$f(x) + C$$

👉 Shift the *IV* graph down by £0.90

At the strike price, **NO** profit is realised.

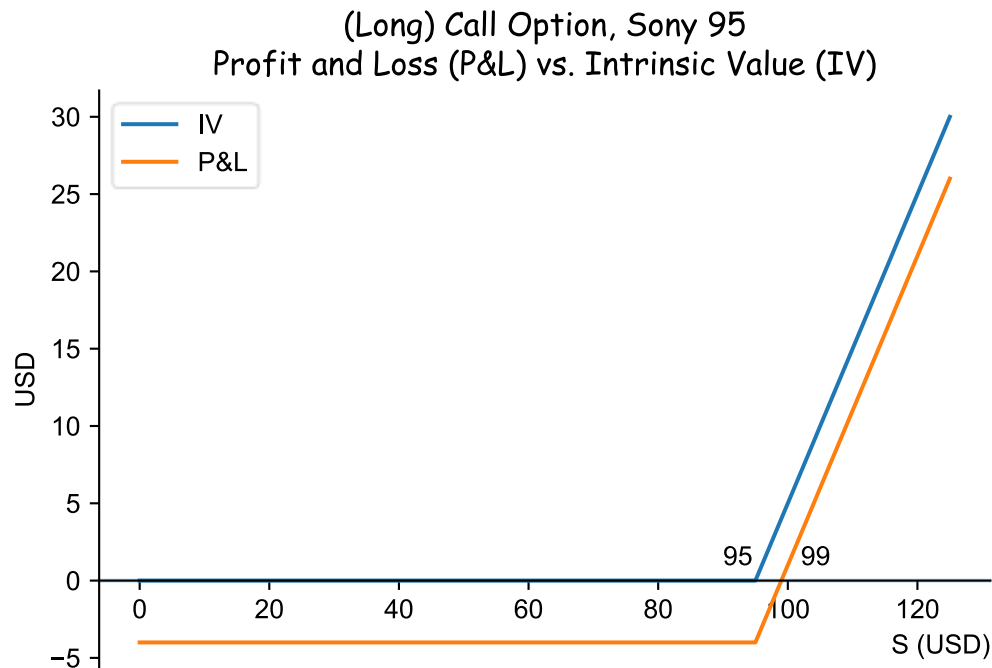
Profit is only realised when $S = 310$ GBX.

P&L graphs – Call Options

Going back to the Sony 95 Call from earlier, assuming a premium of \$4...

👉 You may know that profit = revenue – cost.

👉 Revenue = **payoff**, cost = **premium**.



$$\text{Profit} = IV - 4.00$$

$$f(x) + C$$

👉 Shift the *IV* graph down by \$4.00

At the strike price, **NO** profit is realised.

Profit is only realised when $S = 99$ USD.

Option Strategies

Here's where things really kick up a gear... Now, things are going to be exciting!

☞ Combining options together for specific payoff methods is known as a **strategy**.

☞ The payoff diagrams generated are quite unique.

☞ There are 2 strategies you will be expected to understand...

Short straddle

from a seller's perspective

Long straddle

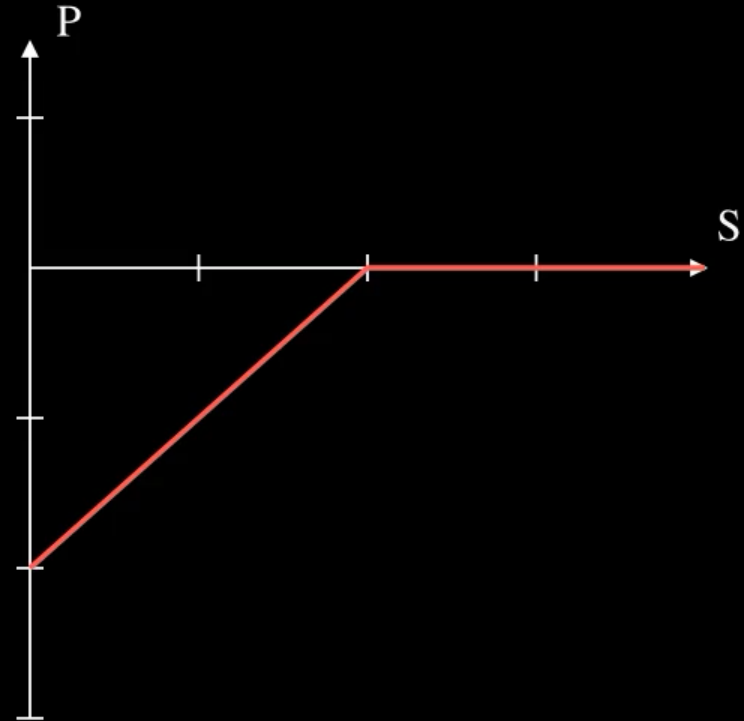
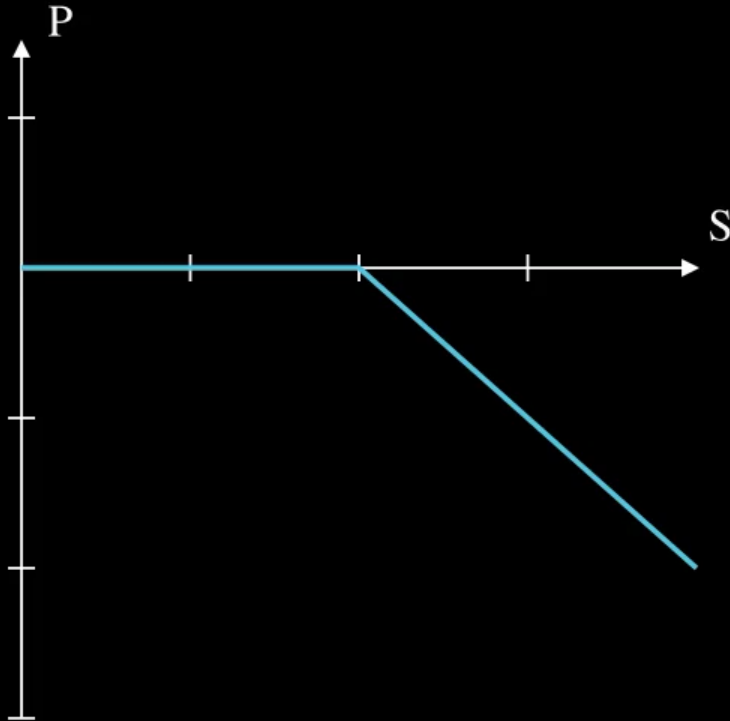
from a buyer's perspective

Your notes

Short Straddle **SELLER'S PERSPECTIVE**

Short Straddle **SELLER'S PERSPECTIVE**

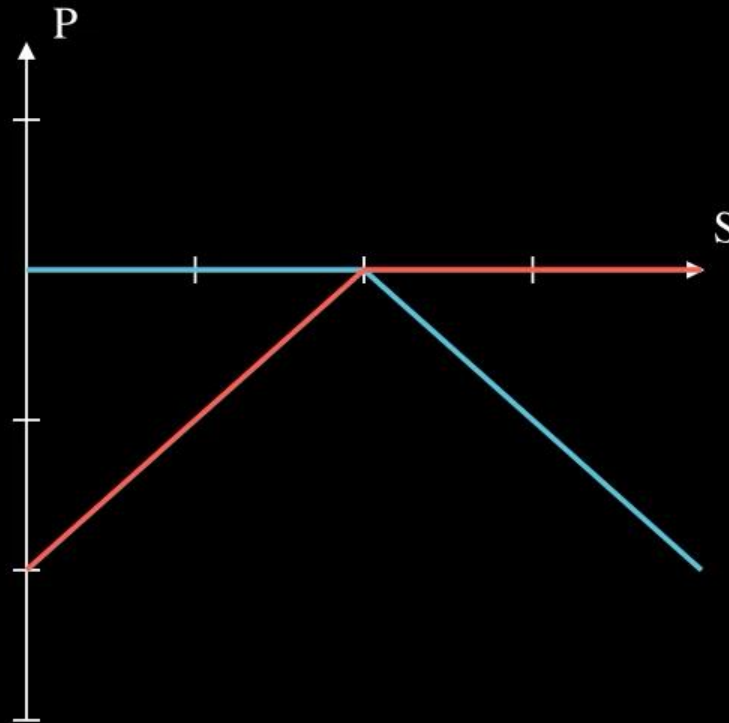
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Short Straddle **SELLER'S PERSPECTIVE**

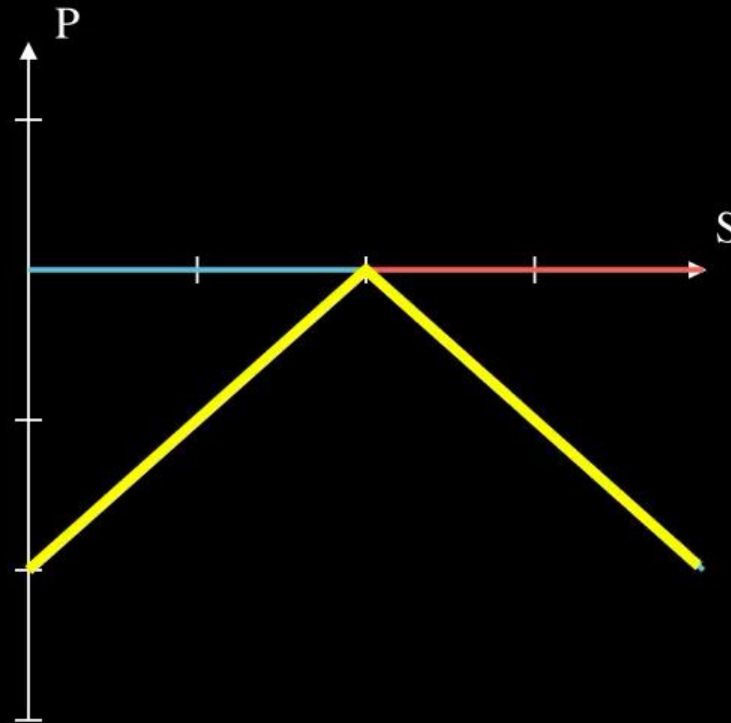
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Short Straddle **SELLER'S PERSPECTIVE**

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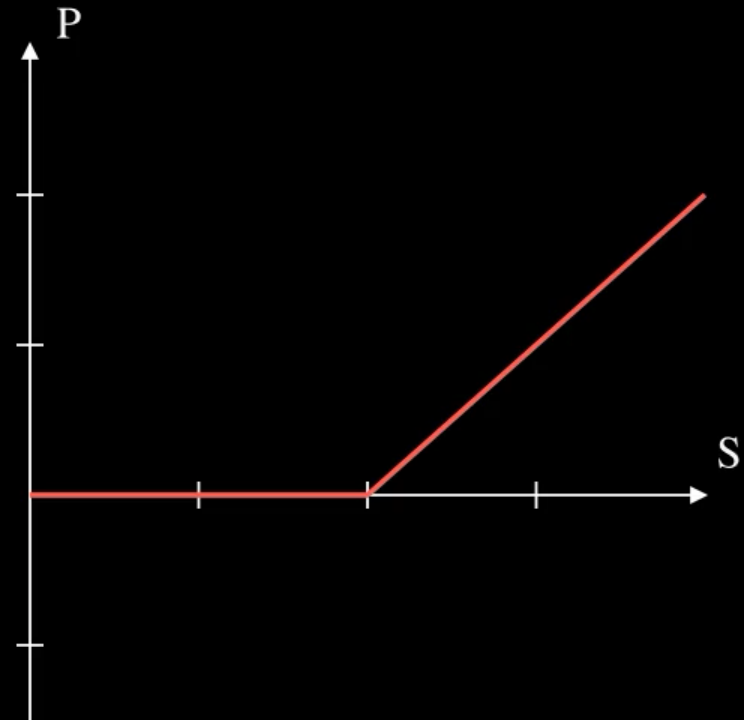
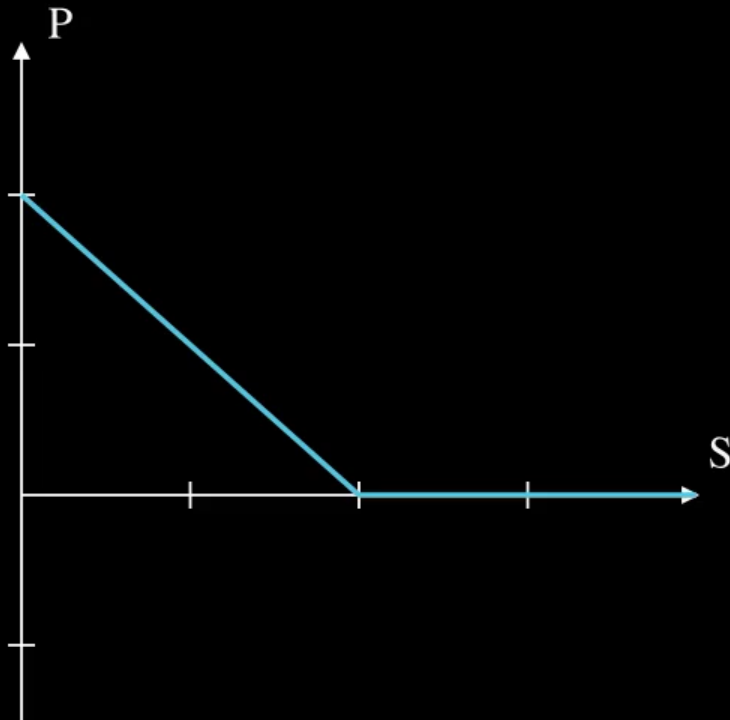


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Long Straddle **BUYER'S PERSPECTIVE**

Long Straddle **BUYER'S PERSPECTIVE**

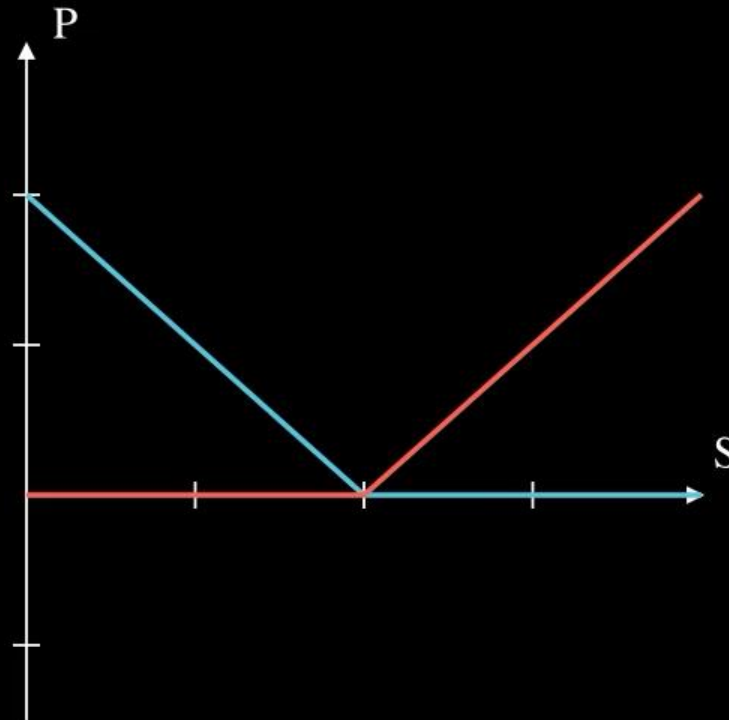
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Long Straddle **BUYER'S PERSPECTIVE**

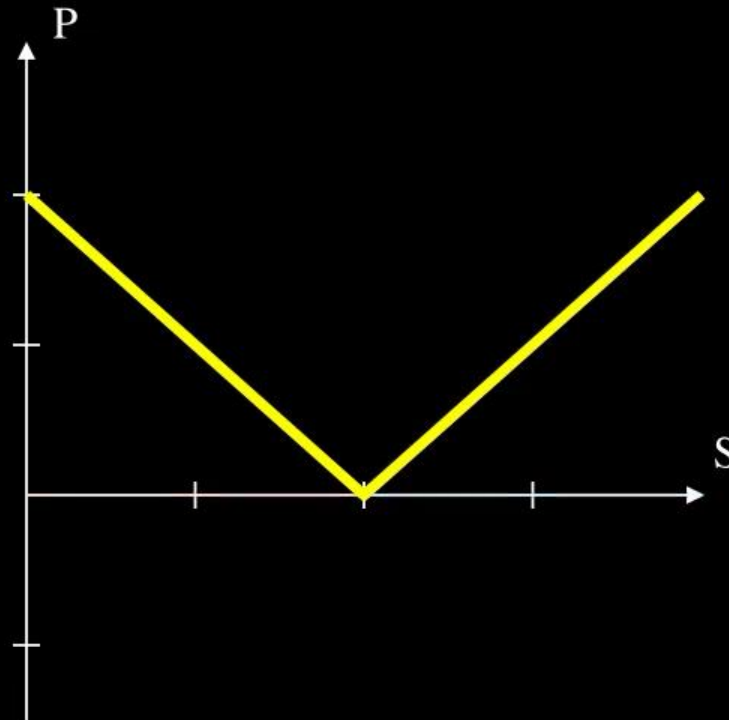
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Long Straddle **BUYER'S PERSPECTIVE**

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Black-Scholes Model

How do traders really calculate the premium of an options contract?

☞ The premium of an options contract is *indirectly* linked to the contract's intrinsic value.

☞ One of the most important equations in derivative pricing is the **Black-Scholes Model**.

☞ There are two types of vanilla options:

European

American

☞ They do share some differences:

Cannot be exercised
until expiry.

Can be exercised
until expiry.

☞ The Black-Scholes model works best primarily with European-style options.

Your notes

Black-Scholes Model

So, let's look at the exact equation of the Black-Scholes model...

👉 The Black-Scholes model is made up of 3 distinct parts:

$$C(S, t) = N(d_1)S - N(d_2)Ke^{-rt}$$

$$d_1 = \frac{1}{\sigma\sqrt{t}} \left[\ln\left(\frac{S}{K}\right) + t\left(r + \frac{\sigma^2}{2}\right) \right]$$

$$d_2 = \frac{1}{\sigma\sqrt{t}} \left[\ln\left(\frac{S}{K}\right) + t\left(r - \frac{\sigma^2}{2}\right) \right]$$

C is the European Call
Option Premium (or **C**ost)

S is the stock price
i.e. the price of the UL.

K is the strike price

r is the risk-free
interest rate

t is the time to expiration

σ is the standard deviation of
the option's previous returns.

N is the normal distribution.

Black-Scholes Model

You've seen this equation, er... indirectly.

☞ The Black-Scholes model is identical to the equation for a call option's intrinsic value, IV.

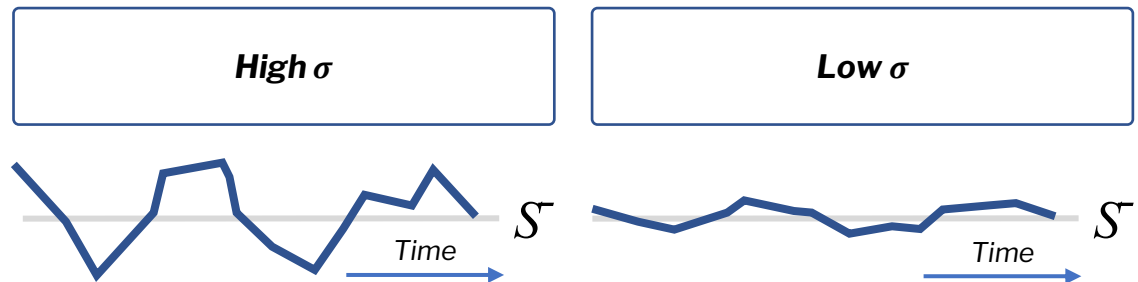
$$C(S, t) = N(d_1)S - N(d_2)Ke^{-rt}$$

$$IV = S - K$$

☞ The Black-Scholes model weights the stock price S and the strike price K for the call premium, C .

☞ σ measures the standard deviation of the UL's price, S .

☞ Visually, this can be illustrated:



N.B. $\bar{S}$ means the mean of S (c.f. arithmetic mean of x is $\bar{x}$)

Black-Scholes Model

How does the Black-Scholes Model work to price an option effectively?

Black-Scholes Model

$$C(S, t) = N(d_1)S - N(d_2)Ke^{-rt}$$

$$d_1 = \frac{1}{\sigma\sqrt{t}} \left[\ln\left(\frac{S}{K}\right) + t\left(r + \frac{\sigma^2}{2}\right) \right]$$

$$d_2 = \frac{1}{\sigma\sqrt{t}} \left[\ln\left(\frac{S}{K}\right) + t\left(r - \frac{\sigma^2}{2}\right) \right]$$

☞ If $S \gg K$, C is high. ($\gg$ means 'much greater')

☞ $\ln\left(\frac{S}{K}\right)$ is large, so d_1 and d_2 increase.

☞ $N(d_1)$ & $N(d_2) \sim 1$ i.e. high likelihood of payoff.

☞ C is high, as a result.

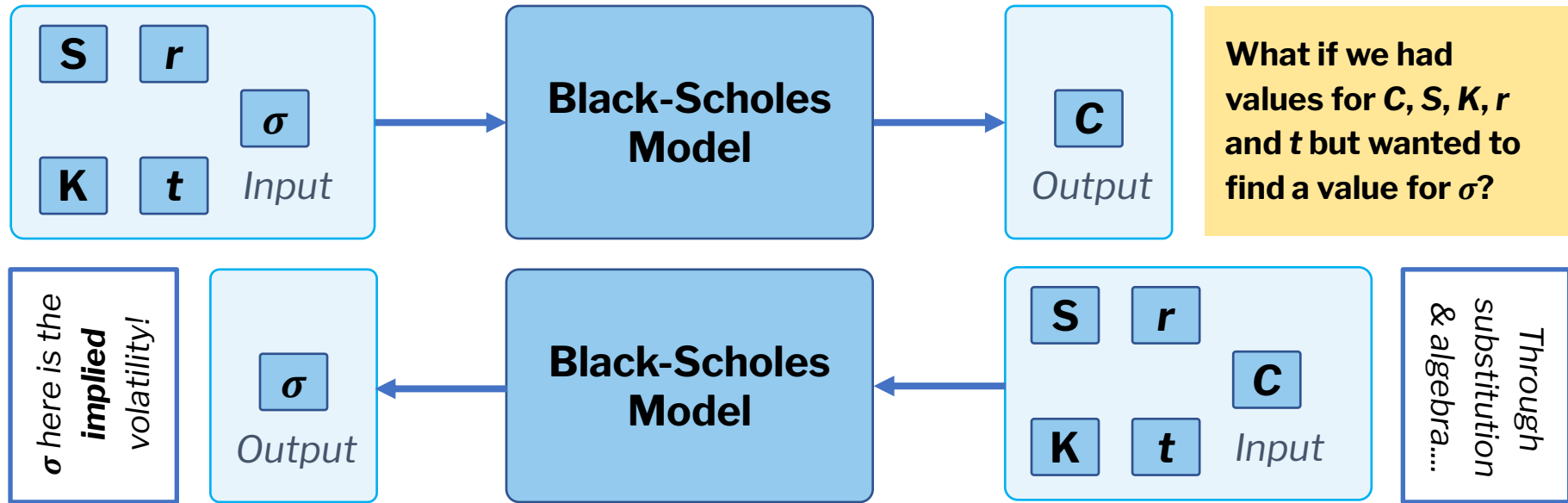
☞ t is time to expiration, so e^{-rt} is the discount factor. If t is small, e^{-rt} is large, so $C \rightarrow 0$.

☞ σ affects $\pm \frac{\sigma^2}{2}$ so if σ is large, d_1 is large but d_2 is small $\rightarrow C = S - K$.

Implied Volatility

You'll hear about **implied** volatility, but where does it come from? The BSM!

👉 The Black-Scholes Model takes 5 unique inputs to output the option premium, C .



👉 No information about future prices is given.

👉 Only the *theoretical* volatility is given.

Issues with the Black-Scholes Model

The Black-Scholes model isn't perfect or reliable to use all the time... Why?

☞ The Black-Scholes model is a solution to a Partial Differential Equation (PDE):

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS \frac{\partial C}{\partial S} = rC$$

☞ Derived from random motion i.e. **Brownian Motion**

1. It is assumed that r and σ are constant → unlikely in the real world.
2. σ uses past historical data → '*past performance does not indicate future results*' (e.g. GameStop).
3. The Black-Scholes model only applies to European Call Options → limited in potential applications.

Your notes

The Greeks

This is a complicated, but common interview question... so, let's look at them!

☞ As the BSM is a solution to a PDE, derivatives can be identified.

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS \frac{\partial C}{\partial S} = rC$$

Delta	Δ	$\frac{\partial C}{\partial S}$
-------	----------	---------------------------------

Gamma	Γ	$\frac{\partial^2 C}{\partial S^2}$
-------	----------	-------------------------------------

Theta	θ	$\frac{\partial C}{\partial t}$
-------	----------	---------------------------------

Vega	V	$\frac{\partial C}{\partial \sigma}$
------	-----	--------------------------------------

Rho	ρ	$\frac{\partial C}{\partial r}$
-----	--------	---------------------------------

The Greeks

☞ Each derivative is named after a Greek letter.

☞ Greeks are used to *hedge* option positions.

☞ They *also* help identify undervalued options.

☞ All Greeks identify a rate of change of C .

A case study example better exemplifies the Greeks in action.

First, a quick dive into each Greek.

The Greeks

What does each Greek mean? How do we read them appropriately?

☞ Delta (Δ) tells you how much the option price C changes if the underlying price S changes by \$1.

☞ Gamma (Γ) tells you how fast the rate of Δ changes.

☞ Γ is the derivative of Δ .

☞ Theta (θ) tells you how much the option price C changes as time passes → this is called **time decay**.

☞ Vega (V) tells you how much the option price C changes if the implied volatility σ changes by 1%.

☞ Rho (ρ) tells you how much the option price C changes if the interest rate increases by 1%.

Your notes

The Greeks – Case Study

Consider a European Call Option with the following parameters:

S = \$10

C = \$1.00

$\Delta = 0.40$

$\Gamma = 0.10$

$\theta = 0.20$

V = 0.04

👉 If $S \uparrow$ from \$10 $\rightarrow$ \$11, $C \uparrow$ by 0.40 to \$1.40, due to Δ .

👉 If $S \uparrow$ again from \$11 $\rightarrow$ \$12, $\Delta \uparrow$ by 0.10 to 0.50. Then, $C \uparrow$ by 0.50 to \$1.90.

👉 Assuming a constant θ , the option loses \$0.20 of its value every day.

👉 Take σ to be 8%. If $\sigma \uparrow$ to 9%, $C \uparrow$ by 0.04 to \$1.04.

👉 If $\sigma \downarrow$ to 6%, $C \downarrow$ by 0.08 to \$0.92.

Your notes

The Greeks – Final Words

Greeks look notoriously complicated, but they're very easy to understand.

☞ For a European Call, Δ is between 0.0 to 1.0

☞ For a European Put, Δ is between -1.0 to 0.0

☞ Δ is also known as the **price sensitivity** of an option.

☞ $|\Delta|$ is used as a probability of an option finishing ITM.

☞ $|\Delta| = 0.90$ means 90% ITM.

☞ Δ , θ and v change with time.

☞ **Theta decay** show OTM option prices decay faster near expiry.

☞ ρ is redundant as interest rates *rarely change* significantly in the lifetime of an option.

Your notes

For your information



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Lecture 5

Derivatives II

The following slides on swaps are NON-examinable.

They are only supplementary and will **NOT** be lectured.

For this reason, they are not formatted appropriately and are left as reference material.

Swaps

Finally, we will end our derivatives journey with the most common of all, swaps

👉 Similar to options, futures and forwards, Swaps have 3 main criteria:

An agreement to exchange cash flows

Based on different bases

👉 The most common underlying assets include interest rates and currencies.

👉 A base indicates how cashflows are calculated, e.g. using LIBOR + 20 BPs

At fixed **dates** in the future

👉 Payments are typically annual/semi annual.

👉 The date is when the initially agreed price is exchanged for the fixed amount of asset.

👉 **NOTE** that these contracts create an obligation between the two parties to exchange cash flows.

Vanilla Interest Rate Swaps

With underlying principal of over \$400 tn, IRS lead the derivatives market

A vanilla interest rate swaps is an agreement between two parties, a fixed and floating rate payer

Each party agrees to **make interest payments** to the other **on set dates** based on a notional principal **with payments calculated on different bases** (e.g. LIBOR floating rate vs a predetermined fixed rate)

Fixed Rate Payer

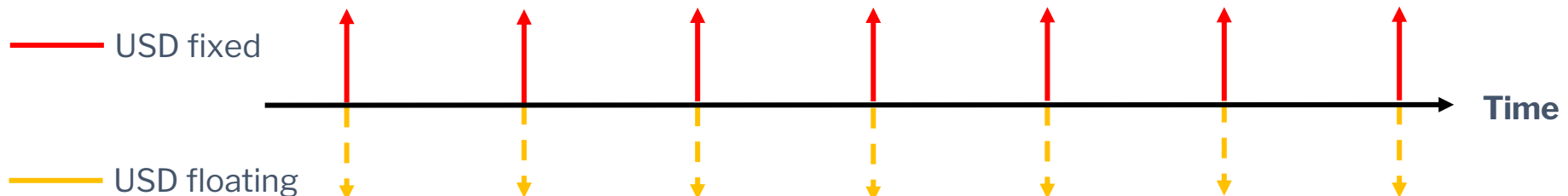
Interest payment at a fixed rate determined at the inception of the swaps, **receives floating rate**

Floating Rate Payer

Interest payments at floating rate determined during the lifetime of the swap, **receives the fixed rate**

IRS have no exchange of principal

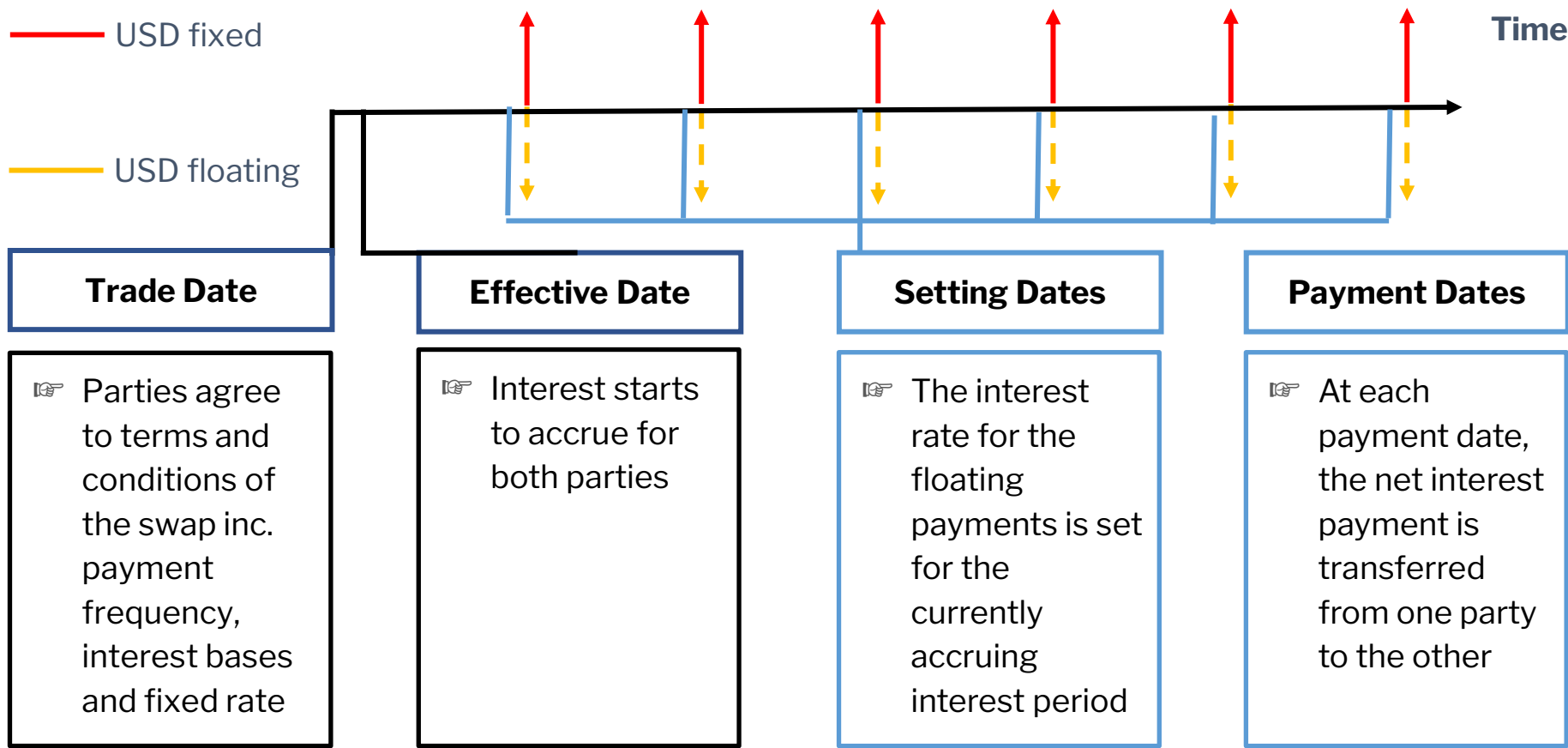
Floating payments usually based on LIBOR or SOFR



Vanilla Interest Rate Swaps

With underlying principal of over \$400 tn, IRS lead the derivatives market

Here the cash flows are shown for the fixed rate receiver – upward arrows indicate income



Interest Rate Swap Example

One of the main applications of IRS is to alter financing costs to fixed or floating

A company, Simperial Ltd., is currently borrowing £10 mn using a bank loan at LIBOR + 1% interest

The company fears that interest rates will increase in future and wants to lock in a fixed rate

How could the company hedge against increasing interest rates over the next 3 years?

They should enter a 3 yr IRS where they receive floating payments and pay fixed.

They enter a swap on £10 mn principal for a fixed rate of 6.5% vs a floating rate of LIBOR + 100 BPs

Similarly to other derivatives, swaps are priced such that no arbitrage opportunities exist. A swap's price considers forward rate curves to ensure that the Present Value of both fixed and floating payments are equal.

Interest Rate Swap Example

Let's look at one scenario which benefits the company if a swap is used

Enter a swap on £10 mn principal for a fixed rate of 6.5% against a floating rate of LIBOR + 100 BPs

Date	Annual Floating Rate	Floating Payment	Fixed Payment	Net Cash Flow
Nov 2022	5.50%	275,000	320,000	- 45,000
Jun 2023	5.75%	287,500	320,000	- 32,500
Nov 2023	5.80%	290,000	320,000	- 30,000
Jun 2024	6.00%	300,000	320,000	- 20,000
Nov 2024	7.00%	350,000	320,000	+ 30,000
Jun 2025	8.00%	400,000	320,000	+ 50,000
Nov 2025	8.50%	425,000	320,000	+ 75,000

Total Profit: + £27,500

Note that interest rates must rise more than the market initially expects for the fixed payer to profit

Currency Swaps

Currency Swaps are like interest swaps, but with an added currency exposure

A currency swap is a two-party agreement

Can be any combination of floating and fixed

☞ Each party contracts to make interest payments to the other on set dates in the future

Fixed/Fixed

Fixed/Floating

Floating only

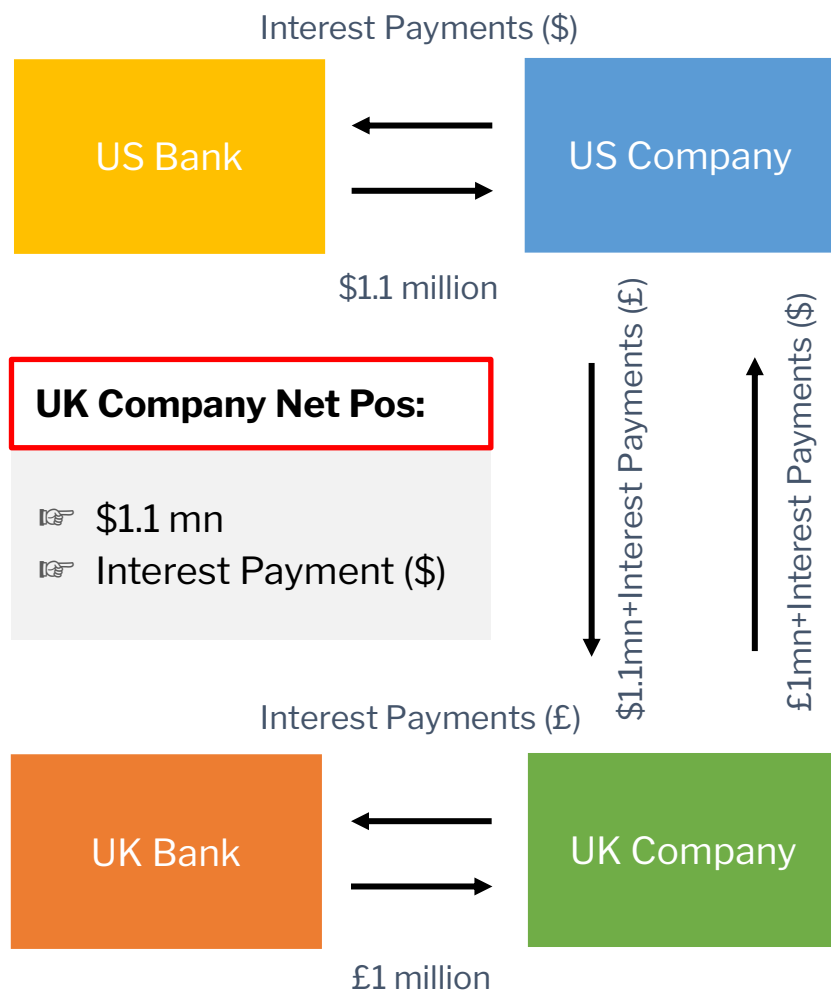
☞ Interest is based on different notional principals in each currency

☞ There is **always** exchange of principal at maturity **at the FX rate at contract inception**

E.g. A company may have debt in a foreign country but may want to pay interest in their own currency

Currency Swaps Example

These contracts are useful when combined with lower domestic interest rates



Companies can get more favorable loans in their own currency than foreign currency

But what if a company needs to borrow in a foreign currency?

They can enter a fixed for fixed currency swap with another company in the target region

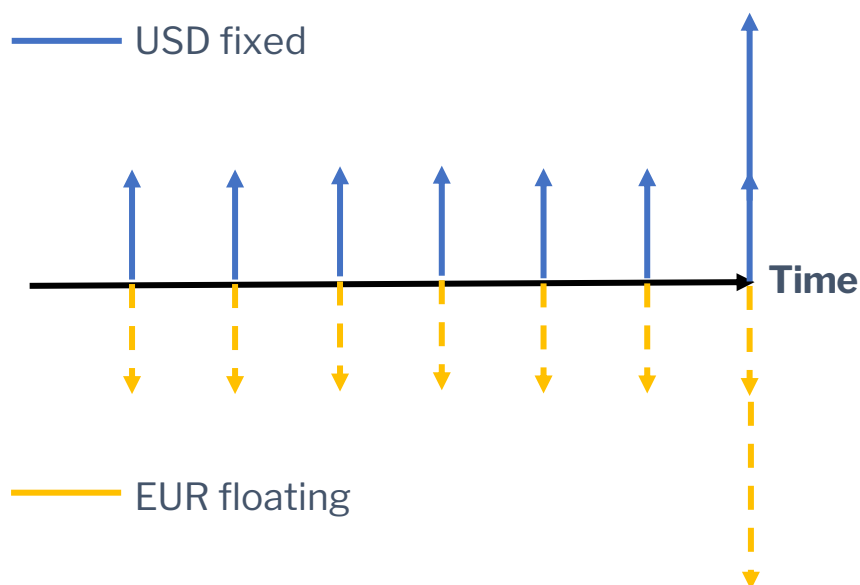
Here imagine that the US company wants to borrow £1 million and the UK company \$1.1 million. Spot: 1.1

Now each company pays the interest rate only available to locals on their foreign debt

Currency Swap Cash Flows

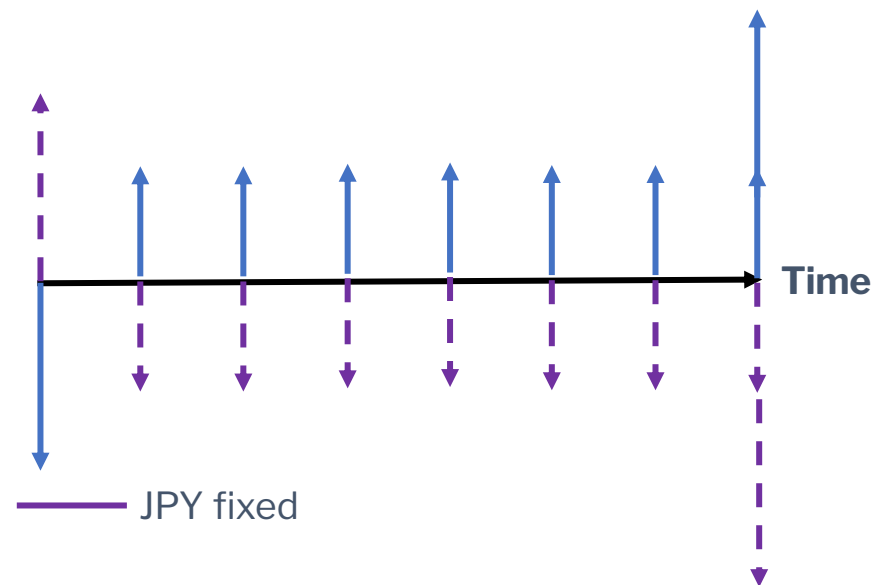
Currency Swaps are like interest swaps, but with an added currency exposure

Both exchanges of principal are executed at the original spot FX rate



Example 1

- ☞ Receive Leg – USD Fixed, Pay Leg – EUR floating
- ☞ No initial exchange of principal



Example 2

- ☞ Receive Leg – USD Fixed, Pay Leg – JPY fixed
- ☞ Initial exchange of principal

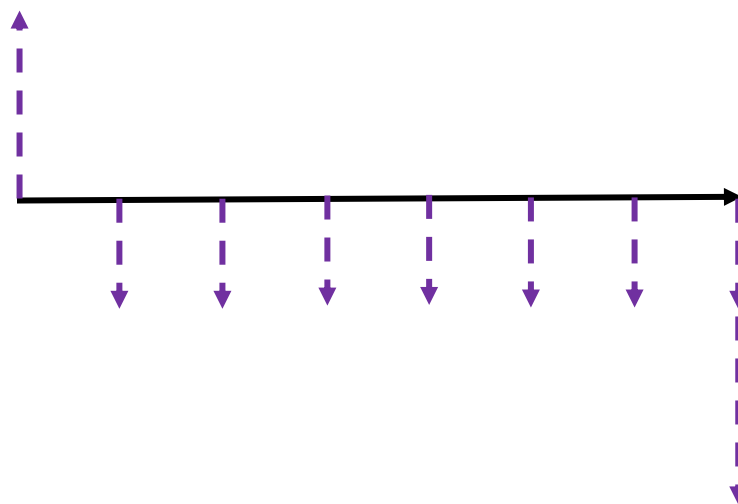
Note that the initial exchanges of principal are in the opposite direction to the final exchange

Currency Swap Example

Currency swaps are often used to convert risky floating payments to fixed

A UK company has access to floating dollar financing (borrowing) but wants to pay fixed sterling

1. The company issues a FRN in dollars



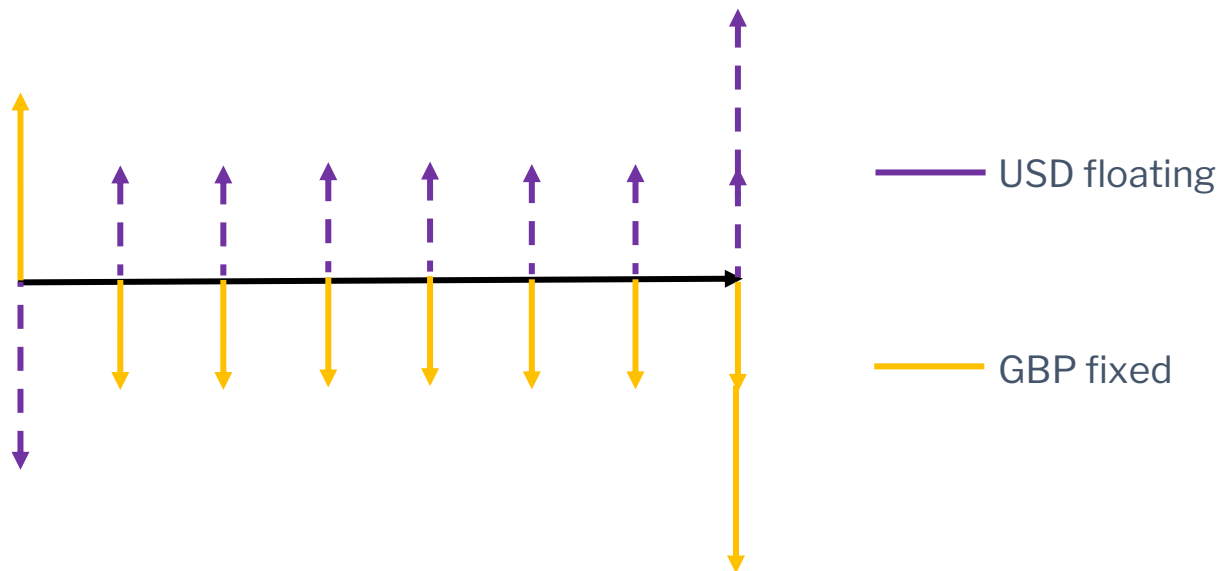
 *FRN – A Fixed Rate Note is a fixed income security with a fixed coupon rate used to borrow from investors*

Currency Swap Example

Currency swaps are often used to convert risky floating payments to fixed

A UK company has access to floating dollar financing (borrowing) but wants to pay fixed sterling

2. Company enters swap paying fixed sterling / receiving floating dollars with initial principal exchange

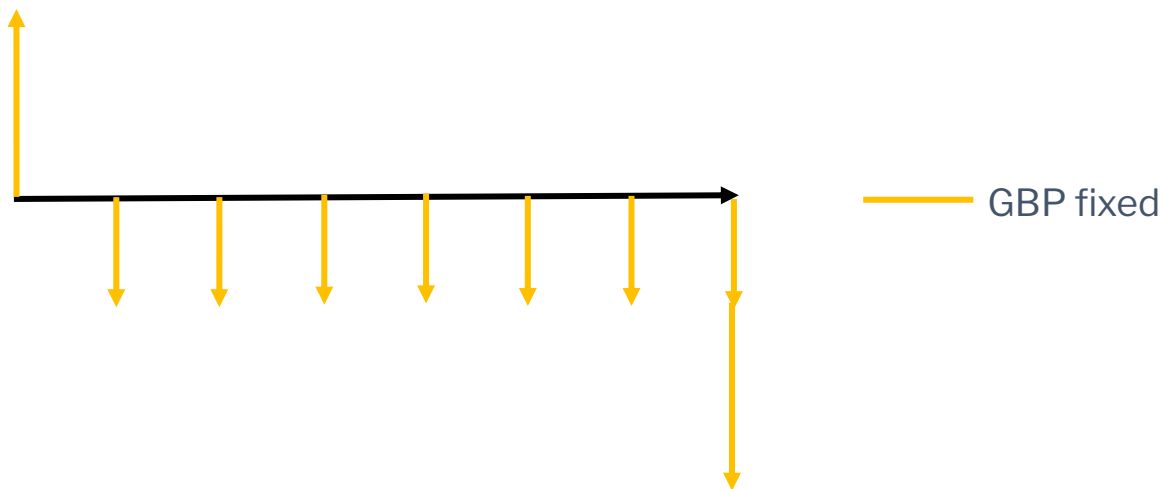


Currency Swap Example

Currency swaps are often used to convert risky floating payments to fixed

A UK company has access to floating dollar financing (borrowing) but wants to pay fixed sterling

3. The cash flows combine to give fixed sterling financing



 Currency swaps enable raising of finance in convenient currencies with preferred currency and basis interest

Credit Default Swaps

CDS behave very differently to the other swaps we have looked at

☞ Provide a kind of **insurance** for the CDS buyer in the case that the underlier **defaults**

☞ The insurance of an underlying debt security (such as a bond) requires scheduled premia.

☞ The CDS seller agrees to pay the face value of the underlying asset in the case of a default

Default

— *Failure of the debt issuer to meet the legal obligations of a loan, e.g., missing a coupon*

☞ The scheduled premia amounts, transaction dates and contract length are specified at inception

In the 2008 financial crisis, these contracts were used by some clever investors to speculate on MBSs and CDOs. When an over-issuance of sub-prime loans led to increasing homeowner defaults nationwide, the underling contracts defaulted and CDS contracts paid out millions.

Why are Swaps so Popular?

Swaps allow banks, corporations, and institutional investors hedge and speculate

Underliers

Interest
Rates

Equity
Prices

Inflation
Rates

Inflation
Rates

Uses of swaps

Hedging

— *e.g. swapping a floating rate for a fixed interest rate*

Asset Swaps

— *The combination of an asset and swap can boost yield or liquidity*

Inflation-linked Swaps

— *e.g. to hedge the liabilities of a pension fund*

Speculation

— *Swaps are commonly used to bet on rates & the yield curve shape*

Vanilla vs Exotic Derivatives

You may have heard us use words such as vanilla and exotic derivatives...

E.g. Vanilla Swaps

Constant Notional Principal

Fixed Swap Rate

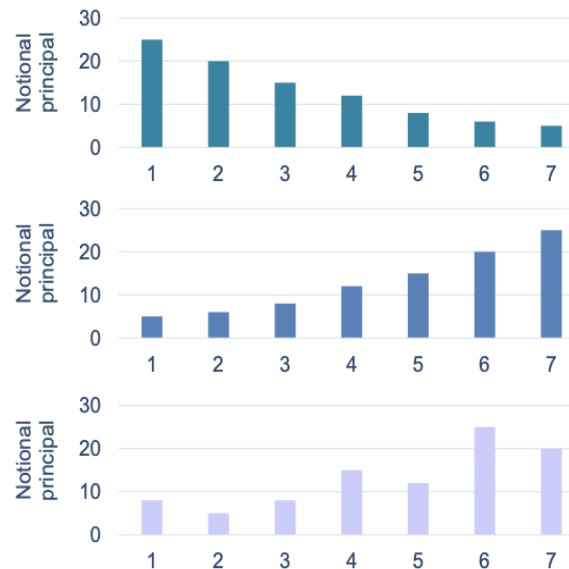
Floating rate set in advance

E.g. Exotic Swaps

Amortizing Swaps

Accreting Swaps

Rollercoaster Swaps



Vanilla derivatives are characterized by fixed terms including time to maturity, strike price and the quantity and quality of the underlying asset. **If any of these terms can change over the life of the contract, it is exotic.** Complex exotics typically have multiple altering terms.